



CS3244 Final Report:

Decoding Gun Violence with Machine Learning

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Group 2

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1. Project Overview

1.1 Objective

The primary objective of this project is to apply machine learning classification techniques to predict the underlying reason for gun-related incidents (Suicide, Homicide, Accidental, Undetermined). This predictive modeling aims to identify key demographic and social risk factors that contribute to different types of violent outcomes in the dataset.

1.2 Dataset

The Guns Incident Data dataset from Kaggle (datatattle, n.d.) contains 100,798 reported gun-related incidents, primarily from Europe, with features such as race, date, education level, and police involvement. Each row represents an incident with the target outcome being the cause (e.g., suicide, homicide). While the dataset is majority Europe-based and compiled from public sources, it offers a large-scale foundation for analyzing patterns in firearm incidents.

1.3 Key Questions / Hypotheses

As part of our project objectives, we focus on the following guiding research questions and/or hypotheses:

1. How do demographic, situational and contextual factors (e.g. race, age, education, place of incident) shape both the reason for gun incidents and the likelihood of police involvement?
2. Are there underlying behavioural or demographic patterns, revealed through clustering, that distinguish different groups of gun incidents? Do these patterns align with the classification outcomes?
3. Can machine learning models reliably identify the key risk factors that differentiate suicides, homicides, accidental shootings and undertermined cases? Do these risk factors reveal broader social or demographic trends observable in the dataset?

2. Data Preparation

This section details the steps taken to clean, impute, and encode the raw gun violence incident dataset, ensuring the data is robust, unbiased, and ready for model learning.

2.1 Missing Value Diagnostics and Assessment

The process began with an exhaustive assessment of missing values to inform the imputation strategy.

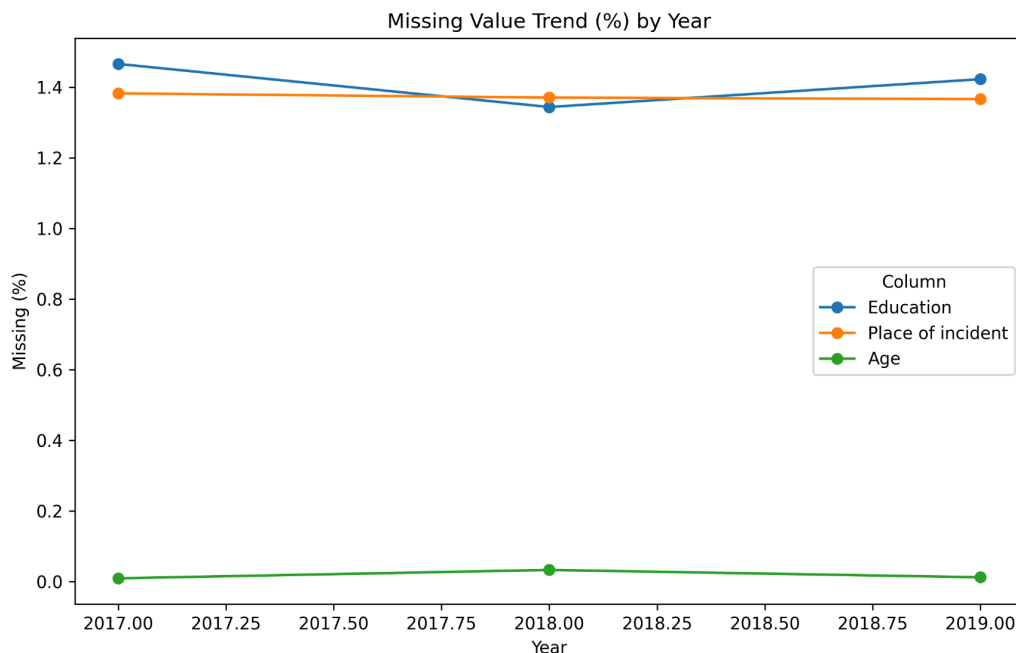
- **Initial Diagnostics**

We identified 3 columns with missing data:

- Education (~ 1.41%)
- Place of incident (~1.37%)
- Age (~0.02%)

- **Missingness Mechanism Analysis:**

- **Temporal analysis** - We analysed the missing rate over the years (2017-2019). The stable and flat trend indicated that missingness was not tied to collection year, strengthening the assumption of Missing At Random (MAR) or Missing Completely At Random.



- **Missingness by Race** - This was a critical finding. The missing rate for Education and Place of incident was found to vary significantly across racial

groups (e.g. Education missingness for Asian/Pacific Islanders was more than double that of Black individuals)

```
Education Missing % by Race:

Race
Asian/Pacific Islander      2.714932
Hispanic                    1.573930
Native American/Native Alaskan 1.526718
White                       1.447831
Black                       1.163290
Name: Education_missing, dtype: float64

Place of Incident Missing % by Race:

Race
Hispanic                    3.103525
Native American/Native Alaskan 2.726281
Asian/Pacific Islander      2.262443
Black                       1.515282
White                       1.050772
Name: Place_missing, dtype: float64
```

- **Conclusion:** This confirmed the data was Missing At Random (MAR), necessitating a group-based imputation strategy rather than a simple global method (Heymans & Twisk, 2022).
- **Correlation:** The correlation between the missingness flags of the three columns was near zero ($r < 0.1$), confirming that while missingness was related to Race (MAR), the features themselves were independently informative.

Correlation between Missingness:			
	Age_missing	Education_missing	Place_missing
Age_missing	1.000000	0.092836	-0.001577
Education_missing	0.092836	1.000000	0.004681
Place_missing	-0.001577	0.004681	1.000000

2.2 Missing Value Imputation

Based on the MAR finding tied to the Race feature, a group-based imputation strategy was implemented to preserve the statistical relationships within the dataset.

Here are the key points for your "Missing Values Imputation" slide, structured to clearly explain the findings, the technical decision (MAR), and the subsequent strategy.

Key Finding / Rationale	Imputation Strategy
To mitigate bias from MAR and preserve statistical relationships, a group-based imputation was implemented.	This method calculated metrics within each Race group instead of using global values.
Categorical Features (Education, Place of incident)	Imputed using the Mode (most frequent value) calculated within each specific Race Group .
Numerical Feature (Age)	Imputed using the Global Mean due to its negligible missing rate 0.02% and low skewness.

2.3 Feature Engineering and Data Finalisation

With all missing values resolved, the final stage prepared the dataset for modeling.

- **Date Feature Engineering:** The Date column was used to derive two new, potentially valuable features:
 - `day_of_week`: The textual name of the day
 - `isWeekend`: A binary flag (1 if Saturday/Sunday, 0 otherwise)
- **One-Hot Encoding:** All remaining categorical columns (e.g. Reason, Sex, Race) were transformed using scikit-learn's OneHotEncoder. This process converted 6 categorical columns into a total of 35 numerical features
- **Final Output:** The final, cleaned, and encoded dataset had a shape of 100,798 rows and 42 columns and was exported as `guns_encoded.csv`, ready for the modeling stage.

3. Exploratory Data Analysis (EDA)

- Sample size: 100,798 gun incident records
- Location: Unknown
- Time period: January 2017 to January 2020 (37 months)
- Data completeness: No missing values across 14 features
- Data quality: Serial Numbers ([S.No.](#)) contain 100,798 unique values, confirming no duplicate records in the dataset

Note on excluded variables for analysis: The dataset includes a numeric Hispanic variable ranging from 100 to 998, with no documentation on its coding scheme. Hence this feature is excluded to prevent any misinterpretation.

3.1 Univariate Analysis

- **Age:** The age distribution shows a right-skewed pattern with two peaks - primary peak at around 20-25 years and a secondary peak at around 50-55 years. This bimodal graph could suggest two high-risk age groups. (Appendix B, Figure 1)
- **Sex:** The gun incidents are generally male dominated, with 85.7% of gun incident victims are male and 14.3% are female. (Appendix B, Figure 2)
- **Education Level:** High School education (42.6%) represents the largest group, followed by Less than High School (21.7%), Some colleges (21.5%) and Bachelors (12.8%). (Appendix B, Figure 3)
- **Race:** White (65.7%) comprises the majority, followed by Black (23.2%), and smaller proportions seen in Hispanic (9.0%), Asian/Pacific Islander (1.3%) and Native American/Native Alaskan (0.9%). (Appendix B, Figure 4)
- **Place of Incident:** A significant portion of incidents occurred at Home (60.0%), with the other minority locations which include Street, Trade/service area etc. (Appendix B, Figure 5)
- **Reason:** Majority of incidents are due to Suicide (62.7%) and Homicide (34.9%), with a small fraction of incidents are Accidental (1.6%) and Undetermined (0.8%). (Appendix B, Figure 6)
- **Police Involvement:** Amongst gun incidents, there is only 1.4% of the incidents involving the police.

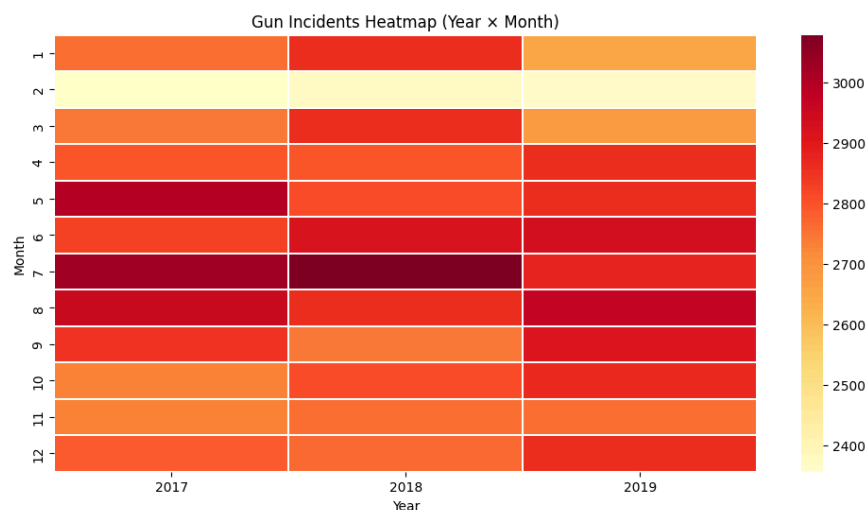
3.2 Temporal Patterns

- The dataset includes gun incidents from January 2017 till January 2020, capturing daily incident counts with considerable variation across days, ranging from around 60 to 145 incidents.
- **Overall trend:** The 30-day moving average reveals a relatively stable incident rate over the mentioned time period, fluctuating around 100 incidents/day with no clear upward or downward trend. This suggests that gun incidents remained at constant levels across 2017 to 2019.

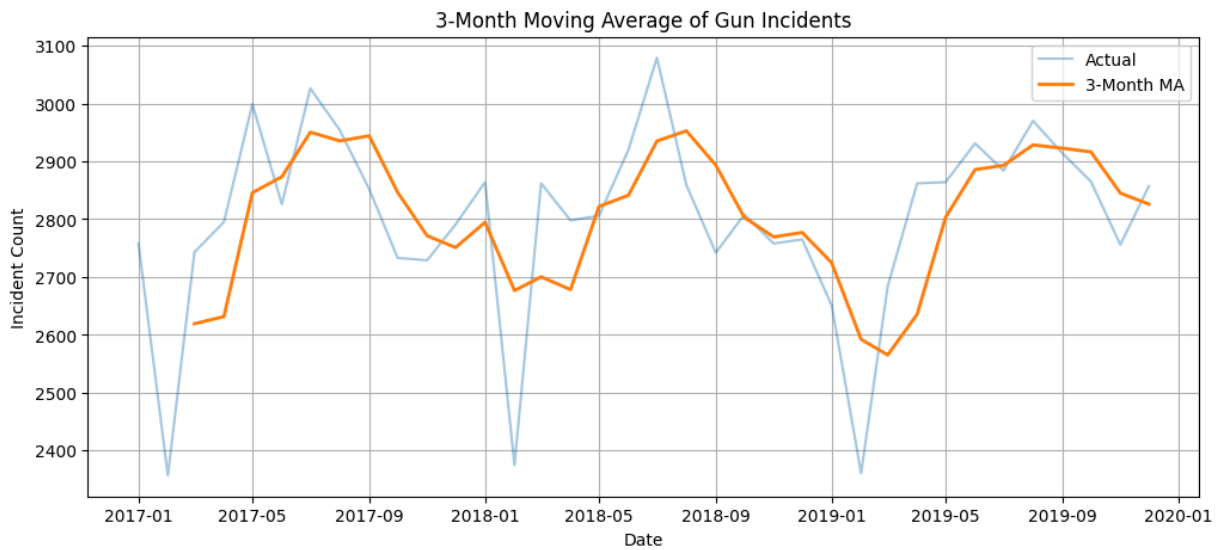
- **Seasonal patterns:** There is a consistent dip in the number of incidents in February every year, likely due to fewer days in February (28-29) compared to the rest of the months (30-31). (Appendix B, Figure 13 & 14)
- **Day of the week patterns:** Incidents are uniformly distributed across the days of the week. (Appendix B, Figure 15)

Time-series analysis was conducted to validate the "Summer Peak" hypothesis.

- **Heatmap Analysis:** The **Year × Month** heatmap reveals a consistent "hot zone" during the middle of the year. Incident counts visibly darken (increase) from **May through August**.



- **Moving Averages:** The 3-month Moving Average (MA) smoothes volatile daily data. The trend shows a cyclical pattern where incidents peak mid-year and trough in February/March.
 - *Statistical Note:* The 3-Month MA often crosses the **2,900+** total incident threshold during peak summer months, whereas winter months frequently dip below **2,500**.



3.3 Bivariate Relationship Against Reason for Incident

- **Age vs Reason:** Age distribution vary significantly across incident types. Homicide victims generally peak at around 20-25 years, while Suicide victims peak at around 55-60 years. Undetermined and Accidental have more uniform distribution across age groups, with slight peaks at around 20-25 years. (Appendix B, Figure 7 & 8)
- **Sex vs Reason:** Predominantly there are more Male fatalities as compared to Female fatalities in each reason for incident. However, this could be due to the fact that there are majority Male as explained above. (Appendix B, Figure 2 & 9)
- **Race vs Reason:** White individuals have at least 4 times more Suicide incidents compared to other incidents which is more than the difference explained in the distribution above (Section 3.1 Reason). Other notable observations include Black individuals having significantly more Homicide incidents. Hispanic individuals also have more Homicide incidents, than other incidents, but not as much as Black individuals, while Asian/Pacific Islander and Native American/Native Alaskan have generally similar types of incidents. (Appendix B, Figure 11)
- **Education vs Reason:** The only significant difference from the Education distribution (Section 3.1 Education Level) is Suicide, where there are lesser fatalities due to Less than High School compared to Some college and Bachelors.

3.4 Bivariate Relationship against Police Involvement

- **Reason vs Police Involvement:** All of the incidents that had police involvement were Homicide incidents. (Appendix B, Figure 16)
- **Sex vs Police Involvement:** A smaller proportion of female fatalities involved police. (Appendix B, Figure 17)

- **Place of Incident vs Police Involvement:** Majority of incidents involving police involvement had Unspecified location, with a small number occurring at Home or Street. (Appendix B, Figure 18)

4. Supervised Learning Models

4.1 Police involvement in gun incidents

Understanding which demographic and contextual factors influence police involvement in gun incidents is important for public safety and equity. Police responses often reflect incident severity, thus identifying key predictors can help target interventions, allocate resources effectively and highlight potential disparities in policing. Using classification models with interpretable feature importance allows data-driven insights to guide policy and prevention strategies (Lee et al., 2024). In this section of the project, our group aims to identify the key factors influencing police involvement in gun incidents.

For this analysis, we treated the prediction of police involvement in gun incidents as a **classification** problem, since the target variable is binary (Yes/No).

4.1.1 Modeling Approach

To capture different patterns in the data and balance interpretability with predictive performance, our group applied 3 modeling techniques, with the following rationales:

Modeling Technique	Rationale / Description
Logistic Regression	Simple and interpretable, allowing us to directly examine the effect of individual features on the likelihood of police involvement.
Random Forest	An ensemble tree-based method that can capture complex nonlinear relationships and interactions between features while remaining relatively interpretable through feature importance measures.
XGBoost	A gradient boosting model that often achieves high predictive accuracy by sequentially correcting errors from previous trees, suitable for handling structured datasets with both categorical and continuous variables.

Subsequently, for a balanced evaluation of the model performance, we used multiple classification metrics:

- Accuracy - Measures the overall proportion of correctly predicted incidents.
- F1-score - Balances precision and recall, especially important if the classes are slightly imbalanced.
- ROC-AUC - Assesses the model's ability to discriminate between police involvement and non-involvement across all classification thresholds.

This multi-model approach allows us to compare simple, interpretable methods with more complex models, ensuring robust insights into the factors that influence police response.

4.1.2 Results Analysis

The performance metrics are summarized below:

===== MODEL PERFORMANCE =====

***Bolded: best performance**

	Accuracy	F1 Score	ROC-AUC
Logistic Regression	0.998810	0.958333	0.999311
Random Forest	0.999802	0.992806	0.993702
XGBoost	0.999504	0.982206	0.998766

All 3 models achieved extremely high accuracy and ROC-AUC, indicating strong predictive performance. Nonetheless, **Random Forest** had the highest F1 Score (0.992806), suggesting it balances precision and recall most effectively among the models, making it most meaningful for our study. While the differences in metrics are small, these results provide confidence that the models can reliably identify whether police are involved in a gun incident.

These results set the stage for interpreting feature importance and understanding which demographic and contextual factors most influence police response.

4.1.3 Feature Importance (Permutation Importance)

To understand which factors most influence police involvement in gun incidents, we examined feature importance using permutation importance on our best-performing model, Random Forest (see above).

In short, permutation importance works by shuffling each feature in turn and measuring the resulting decrease in model performance, allowing us to quantify how much each feature contributes to accurate predictions. Since our dataset is large (~100, 000 rows), as we sampled 1,000 rows, the ranking of the top features usually stays fairly stable. While absolute importance values may vary slightly, the relative ordering of features is generally reliable.

The top 3 most influential features were identified and are summarized below:

Location	Score of importance
Place of incident_Unspecified	0.0133
Reason_Homicide	0.0007
Place of incident_Street	0.0003

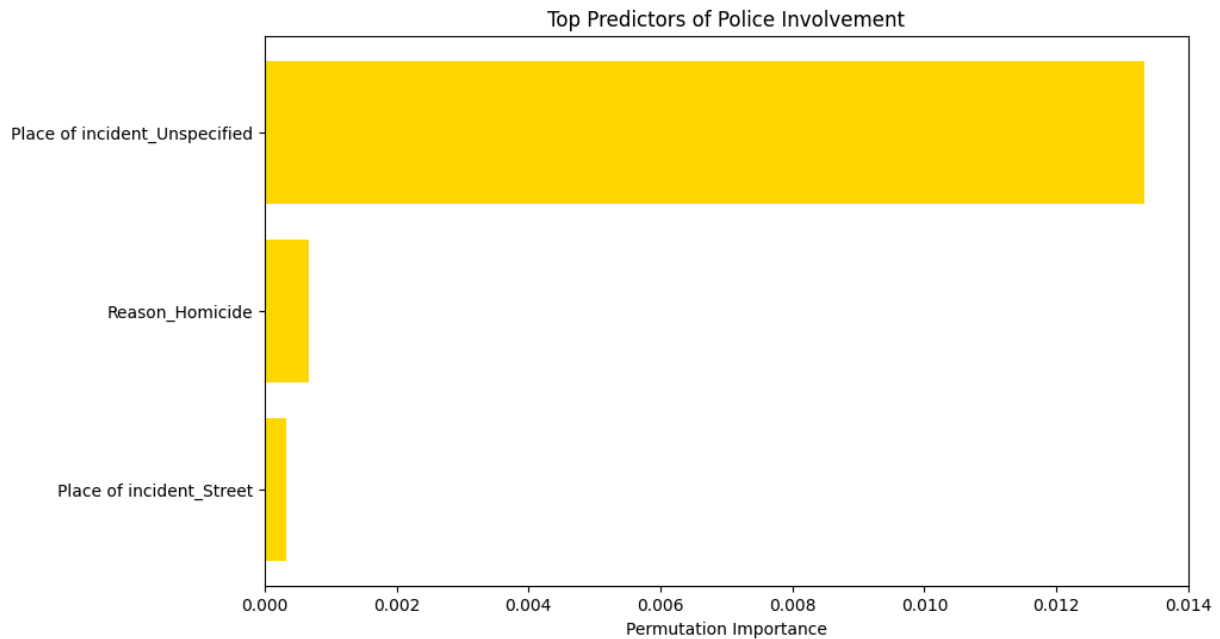


Figure: Top 3 Drivers of Police Involvement

The most influential features point to scenarios where police involvement is more likely, offering insights that can inform resource allocation, public safety strategies and targeted interventions.

Using Random Forest, combined with permutation importance, allows us to capture nonlinear relationships and interactions between features, while maintaining interpretability. These insights will guide the next step of the analysis, error analysis, to examine which subgroups may be under- or over-predicted, ensuring a more equitable and comprehensive understanding of police involvement.

4.1.4 Error Analysis

To better understand where the model may underperform, we conducted an error analysis by examining predictions across different locations of incidents. Specifically, we focused on how well our Random Forest model predicted police involvement in each location category.

The location of a gun incident can significantly influence whether police are involved (Mikdash & Zaiour, 2024). Public areas may prompt faster or more frequent police response due to higher visibility, risk to bystanders or reporting likelihood. Incidents in private residences, on the other hand, might be less likely to be immediately reported or observed. Hence, our group considers the analysis of predictions by location as particularly meaningful, as it not only identifies where the model underperforms, but also uncovers potential systemic biases and

highlights how contextual factors influence police involvement. This insight is critical for ensuring equitable prediction and target public safety interventions.

The analysis was performed by:

1. Creating a test dataset with both true labels and predicted labels.
2. Identifying the most likely “Place of Incident” for each row from the one-hot encoded features.
3. Grouping the data by location and calculating:
 - Total incidents (n)
 - True positives (correctly predicted police involvement)
 - False negatives (missed police involvement)
 - False negative rate (fn_rate)

The resulting table summarizes the model’s performance by location, highlighting areas where predictions are less accurate:

===== FALSE NEGATIVE RATE BY LOCATION =====
*Bolded: Most underpredicted location

location_group	n	false_negatives	fn_rate
Place of incident_Home	12143.0	1.0	1.0
Place of incident_Industrial/construction	58.0	0.0	0.0
Place of incident_Other specified	2790.0	0.0	0.0
Place of incident_Other unspecified	1752.0	2.0	1.0
Place of incident_Residential institution	38.0	0.0	0.0
Place of incident_School/institution	133.0	0.0	0.0
Place of incident_Sports	29.0	0.0	0.0
Place of incident_Street	2256.0	1.0	1.0
Place of incident_Trade/service area	685.0	0.0	0.0
Place of incident_Unspecified	276.0	0.0	0.0

Unspecified locations have a higher false negative rate, indicating that the model is more likely to underpredict police involvement in these settings. For example, while specific locations show minimal or zero false negatives, unspecified locations register a disproportionately higher false negative rate.

This analysis is crucial for understanding potential biases in model predictions and ensuring that high-risk scenarios are not overlooked.

Insights from this error analysis can guide targeted interventions, such as enhancing reporting mechanisms or optimizing resource allocation in locations where police involvement is underpredicted.

Combining this with feature importance results allows for a more equitable and nuanced understanding of the factors driving police response.

4.1.5 Discussion & Conclusion

In this section, we applied 3 classification models - Logistic Regression, Random Forest and XGBoost - to predict police involvement in gun incidents and identify the most influential factors. Random Forest emerged as the best-performing model.

The analyses conducted in this section of the project provide valuable insights into the factors influencing police involvement in gun incidents. Feature importance analysis revealed that **unspecified locations, incidents in the streets** and **homicide-related incidents** as the most influential factors in predicting police response. Error analysis by location further revealed that incidents in **unspecified or underreported settings** are more likely to be underpredicted, highlighting potential areas of systemic bias in model performance.

===== CLASSIFICATION REPORT (BEST MODEL) =====

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19880
1	1.00	0.99	0.99	280
accuracy			1.00	20160
macro avg	1.00	0.99	1.00	20160
weighted avg	1.00	1.00	1.00	20160

Overall, the model demonstrates excellent predictive performance, achieving good precision, recall and F1-scores, which indicates that it reliably captures the majority of police involvement cases. The dataset contains highly informative features - variables that are inherently strong predictors of police involvement - allowing a Random Forest to achieve near-perfect performance. It also suggests the strong, nearly deterministic patterns of the dataset.

These findings have important implications for public safety and policy. Understanding which factors drive police involvement can inform resource allocation, ensuring law enforcement is deployed effectively to high-risk areas. Additionally, recognizing disparities in model predictions emphasizes the need for equitable intervention strategies, particularly in underrepresented or higher-risk settings.

It is important to acknowledge that this study is limited to and relies on available features and historical data. Unobserved factors may also significantly affect police involvement. Future work could explore interactions between features more deeply, include additional contextual variables or apply similar analyses to other outcomes, such as incident severity or

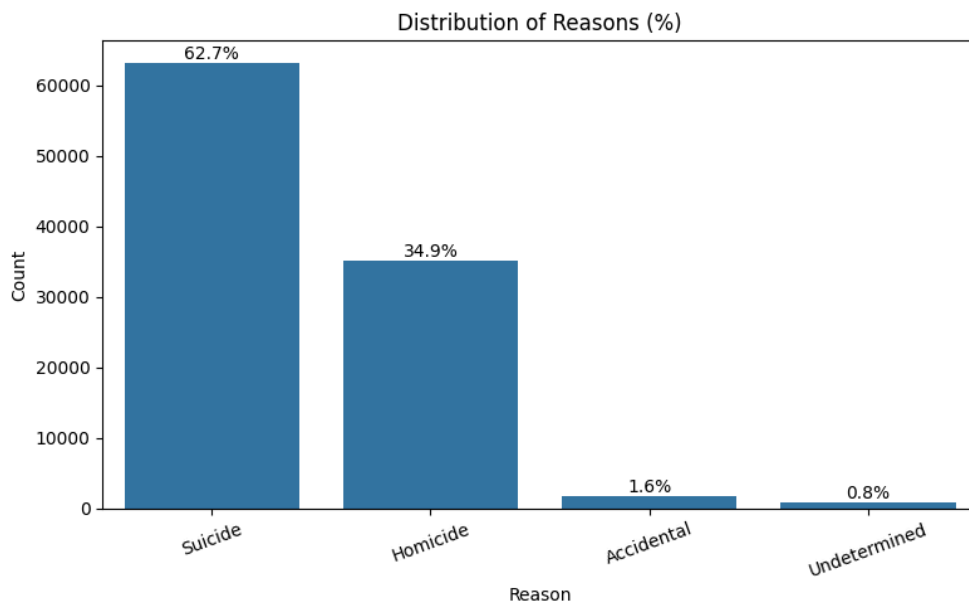
community-level risk. Nevertheless, this analysis is still useful in providing a data-driven foundation to support targeted safety measures and policy planning.

4.2 Classifying reasons for gun incidents

Gun violence has been increasingly alarming, not just in the United States, but also neighbouring countries such as Indonesia, and Malaysia. Studying such data can further highlight the importance of preventive policies, community interventions, as well as risk factor identification. By predicting causes of these gun incidents, we believe this project can provide useful insights into how demographic and social variables influence violent outcomes. Understanding these relationships not only deepens awareness of worldwide gunning incidents, but also helps countries appreciate the value of strong regulation and proactive safety measures to maintain its low rates of gun-related violence.

4.2.1 Addressing the class imbalanced problem

Before going into the modeling, the exploratory data analysis revealed that there is a significant imbalance of classes in the target feature **Reason**.



As shown on the figure, the **Suicide** and **Homicide** classes dominate, taking up around 97.6% of the entire dataset which has 100,798 observations. This leaves the severely underrepresented classes **Accidental** and **Undetermined** taking up only a mere 2.4% of the dataset. Imbalanced classes will result in the model performing poorly on the low frequency classes as it learns more and becomes biased towards the dominant classes. To tackle this issue, sampling methods such as undersampling and oversampling were considered. However, these methods were not implemented because the difference in the number of samples is too huge. Oversampling would risk severe overfitting, while undersampling would

discard too much data. Instead, methods such as tweaking hyperparameters were implemented so as to give more weight to these classes. Data restructuring was also performed to investigate the impact of this issue on our model. This will be discussed further in later sections of the report.

Furthermore, it is important to note that the distribution of gun incident types in our data closely mirrors real-world gun violence statistics, where homicide (~66%) and suicide (~32%) account for the vast majority of cases, and accidental or undetermined deaths occur relatively infrequently (Murthy, 2024). Instead of trying to equalize minority and majority class influence, our task should aim to reflect realistic operational performance. As such, we place more emphasis on micro-averaged scores when we compare between models — which weight classes proportionally to their real occurrence — providing a more meaningful evaluation criterion for this domain.

4.2.2 Modeling approach

To achieve good performance and better insights, 5 different models were implemented. Other than that, 3 baseline models were created which were tuned and refined to improve performance. The method used to refine the models was hyperparameter tuning, where n-fold cross validation was done on the training data. The main metric used in the hyperparameter tuning process was the **micro F1** score. Although macro averaged metrics help to ensure fairness, micro averaging is more realistic given real world frequencies. Additionally, F1 score was chosen over Precision which ignores missed cases and Recall which ignores false alarms. Since the goal is to catch rare incidents while not overpredicting them, F1 score provides a balanced middle ground.

4.2.3 Logistic Regression

(Baseline model)

For the baseline model, the chosen hyperparameters were:

- `max_iterations = 1000`

There are 100,000 over entries as well as many features after one-hot encoding and an imbalanced dataset. This slows optimisation, so a larger maximum number of iterations is required.

- `class_weight = "balanced"`

`class_weight = "balanced"` gives higher importance to minority classes and lower weight to majority ones based on their frequency. This helps to improve precision and recall scores for the underrepresented classes.

- `penalty = "l2"`.

L2 regularisation is the default type of regularisation applied to the model.

Micro F1 score: 0.6436

a. Improving on the baseline

(Logistic Regression L1 regularization)

First, L1 regularization was performed to see if improvements would be observed. More iterations were used because L1 regularisation takes longer to converge. This model did not outperform the previous model.

Micro F1 score: 0.6425

(Logistic Regression L2 with tuned C value)

Next, the C value parameter was tuned to improve performance. The C value affects how strong the regularisation is. Twelve different C values between 10^{-3} and 10^2 were tested. For each C value, 5 fold cross validation was done on the training data and the metric chosen to compare the models was micro F1. The best C value was $C = 0.001$.

```
... Fitting 5 folds for each of 12 candidates, totalling 60 fits

=== New ===
Best params: {'classifier__C': np.float64(0.001)}
Best CV f1_micro: 0.6484
```

Micro F1 score: 0.6527

b. Table of the 3 logistic regression models

This table shows that the tuned L2 logistic regression is the best model as it scores the best for most metrics except for macro Recall and Precision. However, the improvements observed from the tuned model are quite small, with most metrics seeing a less than 0.01 improvement. Overall, improvements on our baseline logistic regression model were made, albeit not by much.

	Accuracy	F1 (Macro)	F1 (Micro)	Precision (Macro)	Recall (Macro)
Model					
Baseline L2 (C=1.0)	0.6436	0.3914	0.6436	0.4325	0.4560
L1 Regularization	0.6425	0.3749	0.6425	0.4177	0.4210
Tuned L2 (C=0.001)	0.6527	0.3940	0.6527	0.4316	0.4517

4.2.4 KNN

(Baseline model)

For the baseline model, the chosen hyperparameters were

- `n_neighbours = 3`

This was chosen randomly

- `metric = "manhattan"`

Manhattan distance was used instead of euclidean distance because the dataset has categorical features and after one hot encoding, these features become binary (0/1). Manhattan distance measures how many binary positions differ — suitable for 0/1 data.

- `weights = "uniform"`

`weights = "uniform"` was chosen because it's simpler and more stable for noisy, real-world data. It also avoids overfitting to the exact numeric scaling of features.

Micro F1 score: 0.7953

a. Improving on the baseline

(Tuning the k value)

To improve on the baseline KNN model, hyperparameter tuning was performed to get the best k value. Twenty different k values, the odd numbers between 1 and 39, were tested. For each k value, 5 fold cross validation was done on the training data and the metric chosen to compare the models was micro F1. The best k value was 21.

Micro F1 score: 0.8296

b. Table of the 2 KNN models

The tuned KNN model performs better in every metric compared to the baseline KNN model, with the micro F1 score increasing from 0.795 to 0.830. Overall, improvement to our baseline KNN model was achieved.

... === Comparison of KNN Models ===

	Accuracy	F1 (Macro)	F1 (Micro)	Precision (Macro)	Recall (Macro)
Model					
Baseline KNN (k=3)	0.7953	0.4087	0.7953	0.4100	0.4079
Tuned KNN (k=21)	0.8296	0.4108	0.8296	0.4124	0.4113



4.2.5 Tree-based models

(Baseline model)

The baseline model implemented was a regular decision tree model.

For the baseline model, the chosen parameters were:

- criterion = “entropy”

criterion = “entropy” uses information gain to decide which feature to split on at each node.

- max_depth = None

max_depth = None means there is no limit how deep the tree can grow to

- class_weight = “balanced”

class_weight = “balanced” gives higher importance to minority classes and lower weight to majority ones based on their frequency. This helps to improve precision and recall scores for the extremely rare classes.

Micro F1 score: 0.6617

a. Improving on the baseline

To improve on the baseline decision tree model, two ensemble learning methods were implemented, Random Forest and XGBoost. These models improve predictive performance by combining the predictions of multiple decision trees.

i. Tuned Random Forest

The random forest model was tuned by testing different combinations of the number of trees used and the maximum depth of the trees. For each combination, 3 fold cross validation was performed on the training data and the metric chosen to compare the models was micro F1. The combination that produced the highest micro F1 score was 200 trees with a maximum depth of 20.

```
Fitting 3 folds for each of 20 candidates, totalling 60 fits
/usr/local/lib/python3.12/dist-packages/joblib/externals/loky/process_executor.py
warnings.warn(
Best params: {'classifier__max_depth': 20, 'classifier__n_estimators': 200}
Best CV micro-F1: 0.7423
```

Micro F1 score: 0.7309

ii. Tuned XGBoost

The XGBoost model was tuned by testing different combinations of the number of trees, maximum depth and learning rate. For each combination, 3 fold cross validation was performed on the training data and the metric chosen to compare the models was micro F1. The combination that produced the highest micro F1 score was 100 trees with a maximum depth of 6 and a learning rate of 0.1.

```
... Fitting 3 folds for each of 20 candidates, totalling 60 fits
Best params: {'classifier__n_estimators': 100, 'classifier__max_depth': 6, 'classifier__learning_rate': 0.1}
Best CV micro-F1: 0.8354
```

Micro F1 score: 0.8339

b. Table of the 3 tree-based models

From the table, it can be seen that while the Random Forest displayed improvements from the baseline model, the tuned XGBoost model is actually the best tree-based model as it performs the best for every evaluation metric considered, notably showing a big improvement in micro F1 score, which is what we prioritised. Overall, improvements on the baseline decision tree model were made.

... === Comparison of Tree-Based Models ===

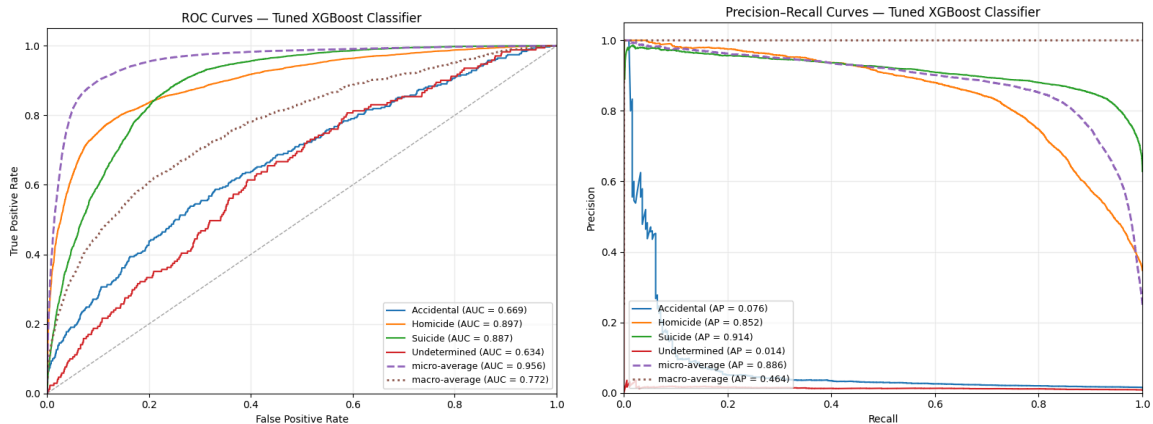
	Accuracy	F1 (Macro)	F1 (Micro)	Precision (Macro)	Recall (Macro)
Model					
Decision Tree (baseline)	0.6617	0.3755	0.6617	0.3975	0.3943
Random Forest (tuned)	0.7309	0.3997	0.7309	0.4106	0.4096
XGBoost (tuned)	0.8339	0.4257	0.8339	0.5393	0.4206

4.2.6 Evaluation

... == Comparison of Tuned Models ==

	Accuracy	F1 (Macro)	F1 (Micro)	Precision (Macro)	Recall (Macro)
Model					
KNN (tuned)	0.8296	0.4108	0.8296	0.4124	0.4113
Logistic Regression (tuned)	0.6527	0.3940	0.6527	0.4316	0.4517
Decision Tree (baseline)	0.6617	0.3755	0.6617	0.3975	0.3943
Random Forest (tuned)	0.7309	0.3997	0.7309	0.4106	0.4096
XGBoost (tuned)	0.8339	0.4257	0.8339	0.5393	0.4206

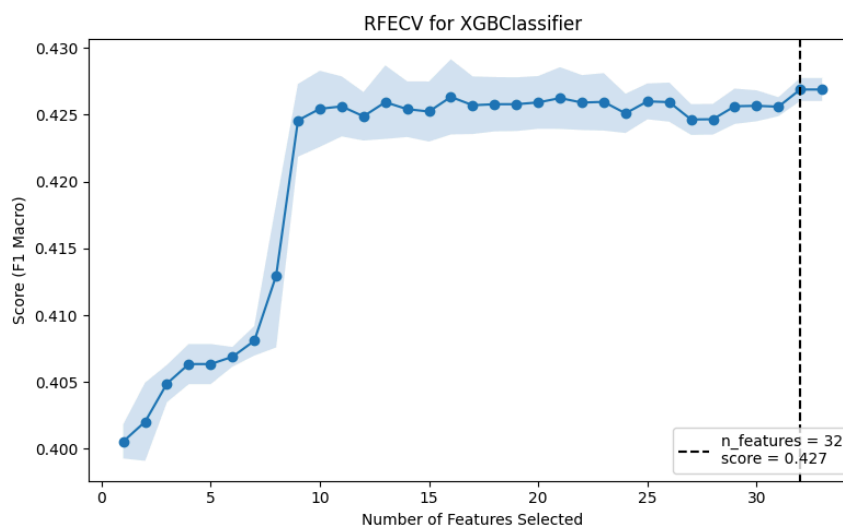
From this table of the five models implemented, it is observed that the tuned XGBoost has the highest scores for all metrics except for macro Recall. This is why XGBoost is chosen as our best model. For the XGBoost model, it has relatively high accuracy and micro Precision, Recall and F1 scores at 0.8220. On the other hand, it does not score well across the macro Precision, Recall and F1 scores, obtaining around 0.42 for macro F1 and Recall as well as 0.55 for macro Precision. This suggests that the model does not do well in predicting the underrepresented classes.



The ROC and PR curve for the XGBoost model further substantiates this point. While the ROC curves for **Accidental** (AUC = 0.67) and **Undetermined** (AUC = 0.63) suggest moderate discriminative ability, the corresponding Precision-Recall curves reveal extremely low Average Precision scores (0.07 and 0.01 respectively). This discrepancy arises because ROC curves are less sensitive to class imbalance, whereas PR curves more accurately reflect the model's poor precision and recall for these rare categories. Therefore, the PR curve provides a more realistic picture of model performance in this imbalanced setting. Overall, poor performance on the underrepresented **Accidental** and **Undetermined** classes is the result of the significantly low frequency of these classes in the dataset.

As mentioned earlier, to address this issue, hyperparameters such as those affecting class weights were chosen carefully to place more emphasis on these extremely rare classes. Nevertheless, it is evident that this did not have a significant impact on the macro averaged metrics. The macro averaged metrics across all models were all around 0.4, with the exception of the chosen XGBoost model where it achieved 0.5 for macro Precision, which is still not desirable.

In order to boost the macro averaged metrics, feature selection was performed by plotting an RFECV to find out if removing features would improve the macro averaged metrics. However, it can be observed from the figure below that the highest macro F1 obtained was only about 0.430 which is still not desirable.



Additionally, data restructuring was done by merging the two severely underrepresented classes together in hopes of improving the macro averaged metrics. Also, to confirm that the low macro scores was due to these two low frequency classes, another restructured dataset where these two classes were completely removed was created as well. To make comparisons fair, all three variants were retrained on a common train–test split with the tuned XGBoost hyperparameters.

=== XGBoost Comparison (Original vs Merged vs Dropped) ===

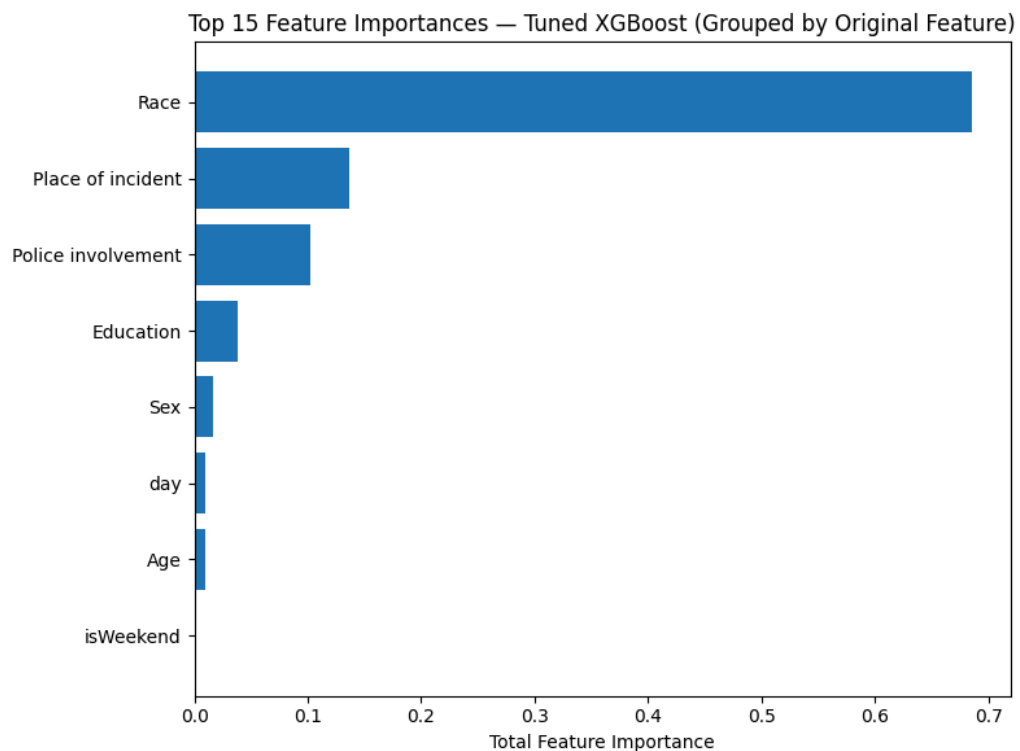
	Accuracy	F1 (Macro)	F1 (Micro)	Precision (Macro)	Recall (Macro)	n_classes	test_size
Model							
XGB — Original (4 classes)	0.8364	0.4278	0.8364	0.5423	0.4216	4	20160
XGB — Merged minorities (3 classes)	0.8372	0.5678	0.8372	0.7405	0.5607	3	20160
XGB — Drop minorities (binary Homicide vs Suicide)	0.8582	0.8405	0.8582	0.8554	0.8308	2	19670

The comparison table clearly demonstrates that the presence of extremely rare classes substantially degrades model performance. When all four classes were retained, the model struggled to learn meaningful patterns for **Accidental** and **Undetermined**, resulting in a low macro-F1 score. Merging these two minority classes into a single “Other” category improved

the macro metrics, indicating that reducing label sparsity helped the model generalise better. However, this merged class still accounted for only about 2.4% of the dataset, meaning the imbalance problem largely persisted and the gains remained limited. In contrast, removing the minority classes entirely produced a marked improvement across all evaluation metrics. This highlights the disproportionate influence that extremely rare categories exert on the model, and reinforces the difficulty of training a multi-class classifier when some target classes are severely underrepresented.

4.2.7 Feature Importance

To understand which variables contributed most to the predictive performance of the tuned XGBoost model, feature importances were computed using XGBoost's built-in `feature_importances_` attribute. XGBoost assigns an importance score to each feature based on the total reduction in model loss (i.e., gain) attributed to splits using that feature across all trees in the ensemble. In other words, features that consistently produce informative splits and reduce prediction error receive higher importance scores.



The resulting plot shows that **Race** is by far the most influential factor in determining the reason for a gun incident, followed by **Police involvement** and **Place of incident**. These three features contribute a majority of the model's predictive power, suggesting that contextual and demographic information strongly shapes the classifier's decision boundaries. In contrast, features such as **Age**, **Sex**, **day**, and **isWeekend** have minimal importance, indicating that they provide little additional discriminatory value in this

classification task. Overall, the plot highlights which predictors the model relies on most and provides insight into where the underlying patterns in the data are concentrated.

The dominance of **Race** and **Police involvement** as top predictors stands out and aligns with broader patterns observed in criminology and public health literature. Racial disparities in firearm-related incidents have been widely documented in the United States, often reflecting underlying socioeconomic inequities, and systemic biases (Jacoby et al., 2018). Similarly, the presence or absence of police involvement provides a strong contextual signal, as law enforcement engagement frequently distinguishes between categories such as homicides, suicides, and other firearm-related events. Place of incident also emerges as a meaningful factor, suggesting that where gun violence occurs—such as in homes, public spaces, or workplaces—carries important information about the nature and motive behind the incident. Conversely, features like **Age**, **Sex**, and temporal markers (e.g., day, isWeekend) appear to play a much smaller role, implying that demographic and time-based characteristics alone are insufficient to differentiate among firearm incident types. These findings collectively indicate that structural and situational factors, rather than individual demographic traits, are more predictive in classifying the reasons behind gun violence incidents in this dataset.

4.2.8 Conclusion and insights

a. Class imbalance is the dominant challenge in creating a fair model

Our model demonstrates that the presence of highly infrequent categories significantly dampened the model's macro-averaged performance. These categories constitute barely ~1% each of the dataset, resulting in extremely sparse class representation. Models trained on such imbalanced data tend to learn the dominant decision boundary well (**Homicide** and **Suicide**) but fail to generalize on rare categories, leading to misleadingly high micro-averaged scores and deceptively poor macro scores. Our work highlights a key takeaway, the challenge in modelling gun violence is not algorithmic, it is statistical. The problem lies in the unequal distribution of different types of gun violence incidents. Until rare categories are better represented, no model, regardless of sophistication or complexity, can reliably learn the causal signals behind those outcomes. This highlights that if our model was to be used in a real world setting, it would only provide value in identifying individuals with a higher likelihood of suicide or homicide risk. Though, its predictions should not be used in isolation. Instead, the model should function as a decision-support tool, as one component within a broader assessment framework that incorporates additional analytical models, professional judgment, and contextual information.

b. Predictive signals are structural, not demographic

The feature importance plot revealed that features such as **Race**, **Place of Incident** and **Police Involvement** were the most influential predictors while **Age**, **Sex** and day-based features were relatively weak. This suggests firearm motivations are linked more to situational

context and systemic factors than basic demographic traits. This aligns with broader criminological research and validates the interpretability of the model. As such, some preventive strategies could include tightening security in certain areas or conducting mental health outreach programs for certain groups who are at higher risk of suicide or homicide.

c. Further improvements

Further improvements could attempt to boost the micro averaged metrics to achieve higher performance (eg. >90%). This could be done by implementing stacking which involves using a combination of different models to train a final model, blending predictions. Other improvements could include improving the fairness of the model, in other words boosting the performance across the underrepresented classes. This could be done by collecting more data on these underrepresented classes.

4.3 Analyzing different risk profiles

In this section of the project, we explored different combinations of features to best explain high-risk against low-risk incidents.

From the offset, we defined an incident as of high severity if its corresponding reason is either homicide or suicide (CDC, 2020).

```
HighSeverity = 1 if Reason in {Reason_Homicide, Reason_Suicide}
else 0
```

4.3.1 HighSeverity Class Distribution

The target variable is highly imbalanced:

Class	Count
HighSeverity=1	98,351
HighSeverity=0	2,447

Severe cases (Homicide/Suicide) dominate the dataset, representing 97.6% of all incidents. This imbalance must be considered when interpreting model performance.

4.3.2 Linear SVM Results/Performance

Metrics Analysis:

- Overall Accuracy (0.64): This metric is misleading due to the high class imbalance. A model that simply predicted HighSeverity=1 for every case would achieve 97.6% accuracy, making 64% seem poor. However, the true performance is in the per-class metrics.
- Performance on HighSeverity (Class 1 - Majority Class):
 - Precision (0.98), Recall (0.64), F1-score (0.78): The model is very precise when it predicts high-risk (only 2% of its high-risk predictions are wrong), but it misses a significant portion of the true high-risk cases (only catching 64% of them).
- Performance on LowSeverity (Class 0 - Minority Class):
 - Recall (0.56): This is the most significant result. The model correctly identifies 56% of the rare low-risk incidents. In the context of the highly imbalanced data, achieving a high recall for the minority class is the goal when using balanced weights. This means the model is providing meaningful insights into the features that truly characterize low-risk incidents.
 - Precision (0.04): The model is extremely poor at being precise when it predicts

low-risk. This means that 96% of the time it predicts a low-risk incident, it is actually wrong (the incident was high-risk). This is a common trade-off when tuning for minority class recall in imbalanced datasets.

The L-SVM provides the most meaningful feature interpretation for discrimination because it uses a linear decision boundary based on the absolute separability of the classes, and the weights reveal the direction of the influence:

These combinations have the highest positive weights, strongly correlating with Homicide or Suicide:

Rank	Feature	Coefficient (Weight)	Implication
1	Police Involvement	1.33	The strongest predictor for high-risk is the presence of police involvement in the incident
2	Place of Incident: Street	0.62	Incidents occurring in public thoroughfares are highly associated with high-risk (likely homicides)
3/4	Sex: Male and Female	0.40	Both gender features are positively weighted, though slightly higher for Sex_Male. When compared against a baseline of unknown/unspecified, this indicates a strong positive association with high-risk outcomes.
5	Place of Incident: Trade/service area	0.38	Commercial areas (e.g., stores, restaurants) are a key location type for high-risk outcomes.
6-10	Race/Education	0.22-0.30	Several demographic and education factors show a positive association. Incidents involving individuals identified as Race_Asian/Pacific Islander, Race_Hispanic, and Race_Black show positive weights, as do individuals with Education_Bachelors.

Summary of High-Risk Profile: High-risk incidents are predominantly characterized by Police Involvement, occurring in public/commercial spaces (Street, Trade/Service Area), and involving individuals across various races and higher education levels.

These combinations have the highest negative weights, strongly correlating with Accidental or Undetermined causes:

Rank	Feature	Coefficient (Weight)	Implication
1	Place of Incident: Sports	-0.48	The strongest predictor for low-risk is an incident occurring at a Sports location, highly suggesting an accidental or undetermined cause.
2	Place of Incident: Other Unspecified	-0.24	Incidents classified under this general location category are significantly more likely to be low-risk.
3/4	Industrial/Construction & Farm	≈ -0.08 - -0.06	Locations related to work or land-use are associated with low-risk outcomes, often implying occupational or environmental accidents.

Summary of Low-Risk Profile: Low-risk incidents are most strongly predicted by their Location Type, particularly in non-residential, sometimes occupational, settings such as Sports facilities, Industrial/Construction areas, and Farms.

4.3.3 Gradient Boosting Results

The Gradient Boosting (GB) model exhibits the classic performance profile of an algorithm on a severely imbalanced dataset when its loss function prioritizes overall accuracy, despite having a capability for complex feature interactions.

Metrics Analysis:

- Overall Accuracy (0.98): This is high, but nearly identical to the baseline accuracy (97.6%) if it simply guessed HighSeverity=1 every time.
- Performance on HighSeverity (Class 1 - Majority Class):
 - Recall (1.00): The model correctly identifies almost all of the high-risk incidents.
 - Precision (0.98): The model is very precise.

- Performance on LowSeverity (Class 0 - Minority Class):
 - Recall (0.02): This is extremely poor. The model only correctly identifies 2% of the true low-risk cases, meaning it missed 98% of them. The model has learned that the safest prediction is always HighSeverity=1, and thus it is effectively failing to identify the low-risk population.

The Gradient Boosting model, while poor at separating the classes, highlights the relative importance of the continuous and time-based features, which the L-SVM downplays:

Feature	Importance (GB)
Age	0.594
Place of incident_Other unspecified	0.123
Month	0.034
Year	0.034

- The GB model emphasizes that Age is the single most important non-categorical predictor of incident risk, followed by the general location and the time of year. This suggests that while linear location and demographic factors are predictive, the absolute age of the individual carries the most information overall when complex feature interactions are considered.

4.3.4 Overall Interpretation

In summary, the models delineate two distinct populations:

1. The High-Risk Population: Characterized by social exposure (public/commercial locations) and intervention (police involvement).
2. The Low-Risk Population: Characterized by environmental specificity (activity-based locations like sports/industrial).

The primary limitation remains the extreme class imbalance. The Gradient Boosting model prioritizes overall accuracy by predicting the majority class (High-Risk) nearly 100% of the time, while the Linear SVM model, using balanced weights, achieves a superior recall of 56% for the rare low-risk events. Therefore, the L-SVM provides the most meaningful insight into the features that truly discriminate between the two risk categories.

5. Unsupervised Learning Models

The primary goal of this unsupervised learning section is to use clustering techniques to objectively identify and profile distinct groups of gun incidents based on their demographic and contextual features.

To achieve this, we employed a two-step approach. We began with Hierarchical Clustering Analysis (HCA) as a foundational, initial exploration to confirm the natural clustering structure and determine a reasonable number of core groups. HCA is computationally expensive and less suited for directly handling our mixed-type data (both numerical and categorical) at scale (Pai, 2021).

Therefore, we followed this with a more extensive and robust analysis using K-Prototypes Clustering. This advanced method is specifically designed for mixed-type datasets, enabling us to scale the clustering to the entire dataset for a highly reliable and granular interpretation of the distinct incident profiles (Huang, 1998).

5.1 Hierarchical Clustering Analysis

For this initial, foundational step, HCA was the preferred method because, unlike partition-based methods (like K-means), it is less susceptible to initial centroid selection bias and provides a valuable dendrogram for visualizing the relationships between data points (Hair 2006)..

5.1.1 Sampling and Linkage Computation

Due to the $O(N^2)$ computational intensity of all-pairs distance calculations in Hierarchical Clustering Analysis (Pai, 2021) for a dataset of over 100,798 observations, a representative subsample of the full, encoded dataset was selected to ensure feasibility and reasonable runtime.

- **Subsampling:** A random subset of **5,040 observations**, 5% of the full dataset, was extracted for Hierarchical Clustering Analysis. This subset maintained the distribution of the key categorical and continuous features from the main dataset.
- **Distance Metric:** The **Euclidean distance** (L2 norm) was the proximity measure used. Although the clustering relies on the Ward criterion (which minimizes the increase in squared error), the distance metric on which Ward operates is the **squared Euclidean distance**. This choice is appropriate for the scaled, high-dimensional feature space (Boehmke & Greenwell, n.d.).
- **Linkage Method:** The **Ward's method** was used for linkage computation. Ward's criterion minimizes the total within-cluster variance (or the sum of squared errors), aiming to create clusters that are **compact and spherical**, which often leads to meaningful and interpretable groupings (Boehmke & Greenwell, n.d.).

5.1.2 Cluster Determination and Evaluation

To determine the optimal number of clusters, we examined the dendrogram and calculated the Silhouette Score for various cluster counts. The largest increase in explained variance, combined with the visual separation in the dendrogram, strongly supported partitioning the data at $k=2$. This dual-method approach validated the choice of two primary incident profiles, providing a clear starting point for the subsequent, more detailed K-Prototypes analysis.

- **Dendrogram interpretation:** The dendrogram revealed that the largest vertical line (highest distance) occurred around Distance 125, suggesting a primary structural split into $k = 2$ macro groups

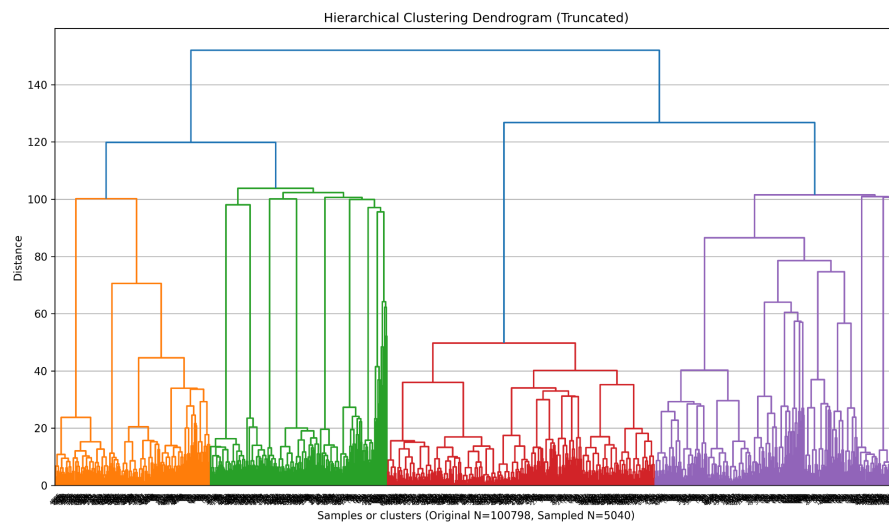


Figure: Hierarchical Clustering Dendrogram

- **Silhouette Score Validation (Step 9):** To confirm this, the Agglomerative Clustering was run for $k = 2, 3, 4$. The **highest Silhouette Score (0.0945)** was achieved at **$k=2$** , quantitatively confirming the fundamental separation suggested by the dendrogram.

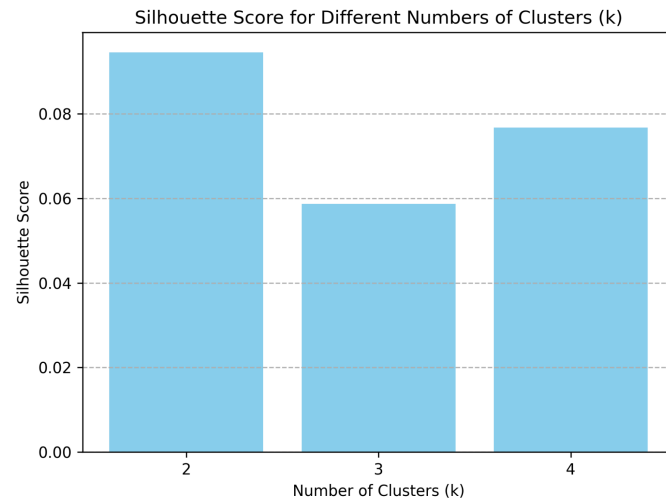


Figure: Silhouette Score for the different k values

- **Final Model:** The model was finalized using **k=2 clusters**, which were visually confirmed to be separated primarily along the **PC2 axis** in the PCA plot.

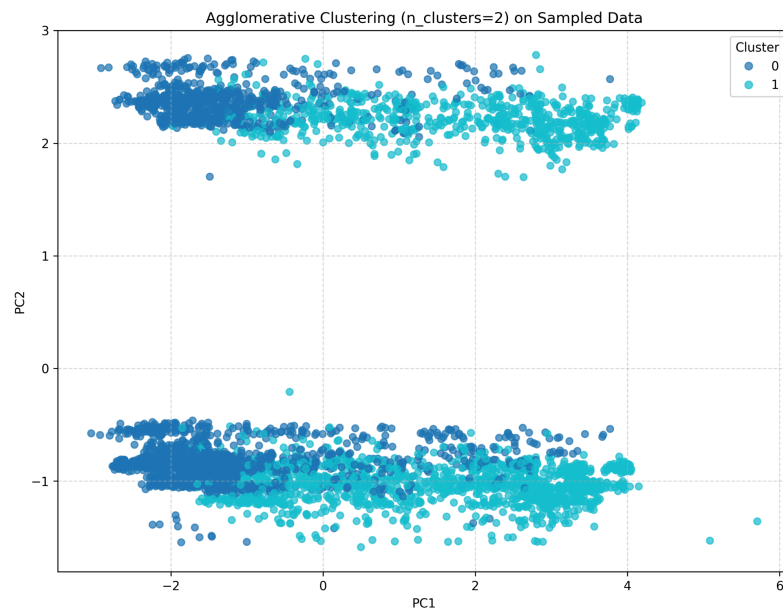


Figure: Agglomerative Clustering for k = 2

5.1.3 Cluster Interpretation

The two final clusters were profiled using the mean of the original features, revealing profoundly distinct characteristics:

Feature	Cluster 0 (The Suicide/Home Cluster)	Cluster 1 (The Homicide/Street Cluster)
Age (mean)	51.34 years	36.46 years
Reason_Suicide	87.5%	31.1%
Reason_Homicide	12.4%	62.8%
Race_White	93.2%	29.9%
Race_Black	6.6%	44.6%
Place of Incident = Home	80.5%	37.2%
Place of Incident = Street	1.0%	23.5%

The Hierarchical Clustering Analysis (HCA) provided a crucial, foundational insight into the binary split of gun incidents into two distinct domains: the “Suicide/Home Cluster” and the “Homicide/Street Cluster”. However, due to the computational intensity of HCA, this simple initial exploration was performed on a subsample of 5,040 observations of the data. Furthermore, effectively handling our mixed categorical and numerical features required complex distance metric manipulation. To overcome these limitations, validate our initial findings on the full dataset, and uncover a more nuanced, age-stratified profiling of incident types, we undertook a more extensive analysis using K-Prototype Clustering.

5.2 K-Prototypes Clustering Analysis

Building directly upon the foundational two-cluster model identified by the Hierarchical Clustering, this section focuses on a more extensive, full-data clustering using the K-Prototypes algorithms. This method was specifically designed by Huang (1998) as an extension of K-Means to handle mixed-type data (both categorical and numerical features) directly and is highly scalable, allowing us to perform the analysis on the entire 100,798 row dataset. This robust approach was implemented to either definitively validate the two core profiles or, more likely, reveal a more granular, age-stratified structure within them.

K-Prototypes algorithm combines both K-Means and K-Modes to handle datasets containing both numerical and categorical variables. The algorithm combines the euclidean distance for numerical features such as Age, and simple matching dissimilarity for categorical features such as Reason for incident, Race and Education etc., which is ideal for our dataset. We aim

to obtain actionable clusters that represent meaningful demographic and behavioural segments within gun violence incidents.

5.2.1 Configuration and Data Preparation

The following features were removed as they either do not contribute to the clustering or do not have much variation in the data.

Features removed

1. **S.No.:** Index
2. **Year:** Minimal variation as seen in the EDA section.
3. **Month:** Minimal variation as seen in the EDA section.
4. **Date:** Cannot process Date-Time feature
5. **Hispanic:** Lack of documentation in the original data source.

The algorithm was conducted on the full dataset of 100,798 observations, with the remaining 35 features (1 numerical, 34 one-hot encoded categorical), preventing loss of information. While K-Prototypes can handle native categorical variables directly without one-hot encoding, we maintained the encoded format to keep constant with the preprocessing steps.

Feature Set (35)

Numerical (1)

1. **Age**

Categorical (35)

1. **Incident type** (4)
2. **Education level** (4)
3. **Sex** (2)
4. **Race** (5)
5. **Place of incident** (10)
6. **Day of week** (7)
7. **isWeekend** (1)
8. **Police involvement** (1)

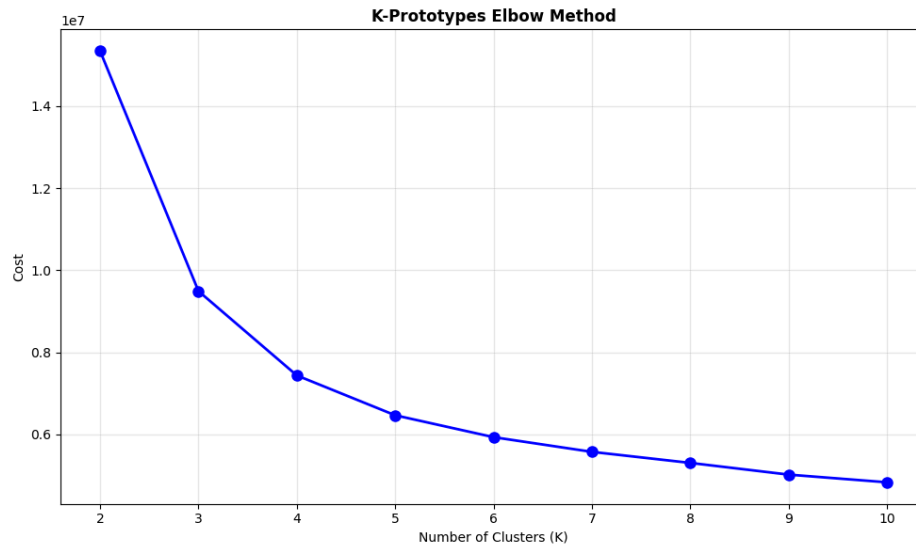
Distance metric: K-Means clustering method is not sufficient as it uses distance metrics such as euclidean distance which is only suitable for calculating distances between numerical features. Hence we employ K-Prototypes, which uses the following combined dissimilarity measure to cluster mixed-type features (Huang, 1998):

$$d(X, Y) = d_{num}(X_{num}, Y_{num}) + g * d_{cat}(X_{cat}, Y_{cat})$$

Where d_{num} is the squared Euclidean distance for numerical features, while d_{cat} is the number of mismatches for categorical features, and g is the weighting parameter that is automatically provided by the algorithm.

5.2.2 Determining the Optimal Number of Clusters

The algorithm tested for $K = 2$ to $K = 11$ clusters, with the optimal number of clusters determined by using the Elbow Method, which maps the **Cost** (total within-cluster dissimilarity) against the **Number of Clusters** K . The lower the Cost, the closer the points are to their respective centroids.



- From the plot above, we see that the cost decreases monotonically, with the rate of improvement slowing down beyond $K = 5$.
- The ideal number of clusters is determined by the “elbow” of the plot.
- The visual “elbow” is observed between $K = 4$ and $K = 5$, suggesting these two values as possible numbers of clusters.
- Hence we shall look into other evaluation metrics to determine the optimal number of clusters.

Comparison between $K = 4$ and $K = 5$ clusters (Geeksforgeeks, 2025):

We see below that $K = 4$ clusters performed better on the majority of the metrics except for Cost, as compared to $K = 5$ clusters.

Metric	K = 4	K = 5	Winner	Interpretation
Cost	~7.43M	~6.46M	K = 5	Lower total dissimilarity
Davies-Bouldin Index	3.66	4.57	K = 4	Lower indicates better cluster separation
Calinski-Harabasz Index	10,800	8,410	K = 4	Higher indicates better-defined

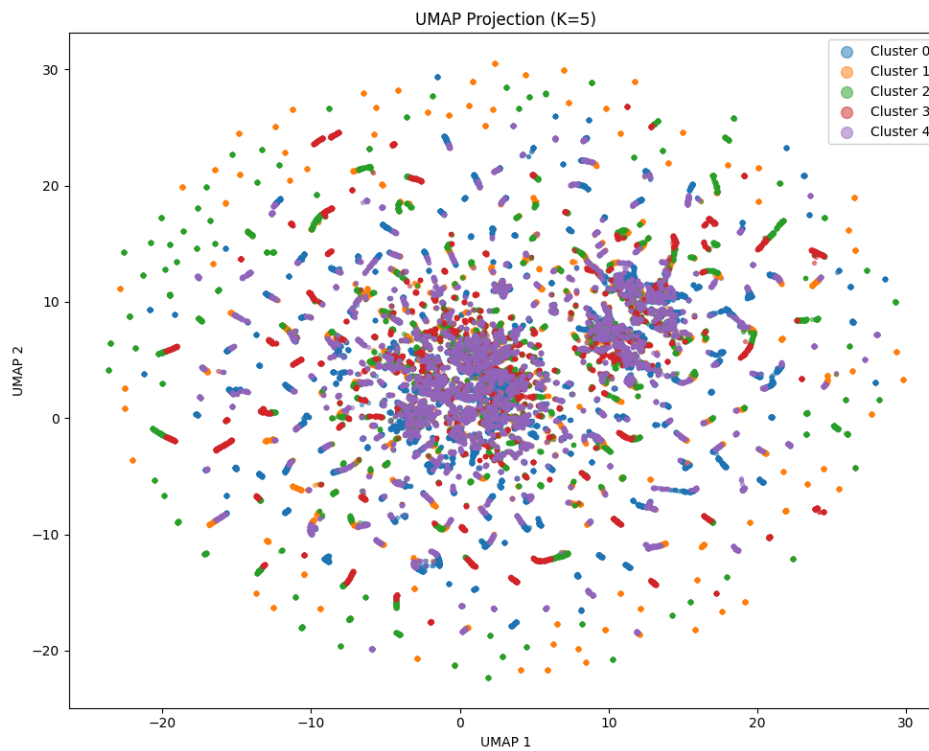
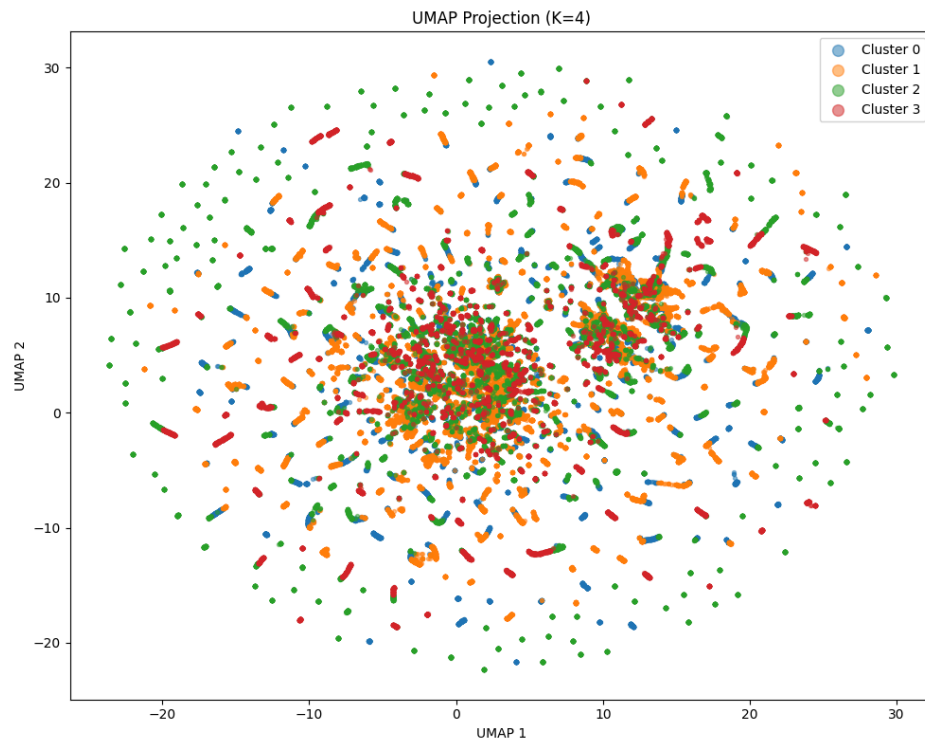
				clusters
Silhouette Score	0.0446	0.0242	K = 4	Measures cluster cohesion and separation

UMAP Projection (K = 4): To visualize the clusters high-dimensional structure, we employ Uniform Approximation and Projection (UMAP), which is similar to traditional dimensionality reduction methods like PCA to transform our data into a lower-dimensional space by identifying axes that results in the greatest variance (Carnot, 2024). In contrast, UMAP is a non-linear method that is particularly well-suited for mixed-type data as it preserves both local and global data structure better than linear methods like PCA that focus only on preserving the global structure of the data (Carnot, 2024).

UMAP has 4 main hyperparameters (UMAP Documentation, n.d.):

- **n_components:** dimension space of map
- **metric:** how distance is calculated between data points
- **n_neighbors:** controls the balance between local vs global structure in the data
- **min_dist:** controls how tightly the points are packed together

We created the UMAP for both K = 4 and K = 5 clusters as seen below.



Both UMAP projections reveal significant overlap between all four clusters, where the clusters converge towards the central region, with several points from different clusters spread further

from the center. There is minimal separation between the two clusters in the two dimensional space. These observations align with the silhouette score of 0.0446 ($K = 4$) and 0.0242 ($K = 5$) which are relatively low (away from 1). Despite the lack of distinct cluster boundaries in both visualizations, it does not mean that the clustering solution is wrong, but rather it reflects the complex, multi-dimensional nature of the data, where meaningful segments appear due to subtle combinations of features rather than large, distinct differences in a single dimension.

Although the overlapping of the clusters in the UMAP is not ideal when trying to obtain distinct clusters, it actually reinforces that gun violence may not be driven by a few dominant features, but rather by subtle interactions between demographic and incident characteristics. The clusters obtained could represent useful demographic and incident segments to understand how gun incidents occur across different factors such as age, race, education and incident circumstances.

5.2.3 Cluster Profiling and Interpretation

We chose the $K = 4$ cluster as it performed better in the majority of the metrics as compared to the $K = 5$ cluster. Hence the $K = 4$ clusters were obtained using both numerical summaries and categorical proportion tables across all the features. The K-Prototypes clustering revealed age-stratified demographic groupings spanning across a human lifespan, with distinct patterns observed in other features such as incident characteristics, demographics and location.

To properly interpret cluster characteristics, we shall compare the cluster profile against the baseline distribution of the data in our analysis of each cluster. For example, gun incidents disproportionately affect 85.7% of male individuals compared to 14.3% of female individuals. Suppose one of the clusters shows a similar gender distribution, then gender may not be a key driver of that cluster, as the proportions may simply reflect the underlying dataset. Alternatively, the similar distribution between the clusters and the baseline distributions could suggest proportional risk exposure across the population demographic.

Cluster Profile from K-Prototypes Clustering

Cluster	0	1	2	3
Size	25,397 (25.2%)	33,301 (33.0%)	27,429 (27.2%)	14,671 (14.6%)
Mean / Median Age	39.9 / 40.0	23.97 / 24.0	57.09 / 57.0	77.97 / 77.0
Race (Top 3)	White (64.1%) Black (23.2%) Hispanic (9.9%)	Black (44.1%) White (38.0%) Hispanic (15.1%)	White (86.0%) Black (8.1%) Hispanic (4.2%)	White (93.5%) Black (3.3%) Hispanic (2.4%)
Sex	Male (83.3%) Female (16.7%)	Male (88.1%) Female (11.9%)	Male (82.8%) Female (17.2%)	Male (89.7%) Female (10.3%)
Education	High School (43.9%) Some College (23.9%) Less than HS (17.5%) Bachelor (13.5%)	High School (41.8%) Less than HS (34.4%) Some College (19.3%) Bachelor (3.6%)	High School (43.0%) Some College (23.9%) Bachelor (19.4%) Less than HS (11.8%)	High School (41.2%) Bachelor (20.5%) Less than HS (18.4%) Some College (18.0%)
Reason	Suicide (60.9%) Homicide (37.1%) Accidental (1.3%)	Homicide (61.2%) Suicide (35.6%) Accidental (2.2%)	Suicide (81.5%) Homicide (16.3%) Accidental (1.5%)	Suicide (92.1%) Homicide (6.3%) Accidental (1.2%)
Location (Top 3)	Home (59.5%) Other specified (14.3%) Street (10.3%)	Home (45.3%) Street (20.7%) Other specified (16.3%)	Home (67.7%) Other specified (12.9%) Other unspecified (8.3%)	Home (80.0%) Other specified (7.8%) Other unspecified (6.7%)
Police Involvement	No (98.1%) Yes (1.9%)	No (98.1%) Yes (1.9%)	No (99.1%) Yes (0.9%)	No (99.8%) Yes (0.2%)
Day of Week (Monday - Sunday)	Uniform distribution (~14.3%/day)	Uniform distribution (~14.3%/day)	Uniform distribution (~14.3%/day)	Uniform distribution (~14.3%/day)
Is Weekend	Weekday (71.3%) Weekend (28.7%)	Weekday (71.0%) Weekend (29.0%)	Weekday (71.3%) Weekend (28.7%)	Weekday (71.0%) Weekend (29.0%)

Population Baseline Data Distribution (PBDD)

Feature	Overall Distribution
Race	White (65.7%), Black (23.1%), Hispanic (9.0%), Asian/Pacific Islander (1.3%), Native American/Native Alaskan (0.9%)
Sex	Male (85.7%), Female (14.3%)
Education	Less than HS (21.7%), High School (42.6%), Some college (21.5%), Bachelors (12.8%)
Reason	Suicide (62.7%), Homicide (34.9%), Accidental (1.6%), Undetermined (0.8%)
Location	Home (60.0%), Other specified (13.6%), Street (11.1%), Other unspecified (8.8%), Trade/service area (3.4%), Unspecified (1.4%), School/institution (0.7%), Farm (0.5%), Industrial/construction (0.2%), Residential institution (0.2%), Sports (0.1%)
Police Involvement	No (98.6%), Yes (1.4%)
Day of Week	~14.3% per day
Weekend	Weekday (71%), Weekend (29%)

Cluster 0: Middle-Aged Adults

- Size: 25,397 Incidents (25.2%)
- Mean Age: 40 Years
- **Demographic Profile:** This cluster closely mirrors the overall PBDD across all features. Race and Education proportions deviate minimally within 2% as compared to the PBDD, while the Sex distribution of females (16.7%) show a slight increase of 2.4% from the PBDD, which is also the second highest female proportion across all clusters. This may potentially reflect domestic violence incidents or female-specific midlife problems.
- **Incident Characteristics:** This cluster shows mixed-motive pattern, with suicide (60.9%), homicide (37.1%) and accidental (1.3%) incidents, which reflects the PBDD with a small deviation of about 2%. Location patterns also depict similar proportions compared to the PBDD, with the top 3 being home (59.5%), other specified (14.3%) and street (10.3%).
- **Interpretation:** Cluster 0 captures adults experiencing life's pressure peak such as managing careers, raising children or supporting aging parents. The close alignment with the PBDD indicates that gun violence affects the different demographic groups proportionally. This represents the typical gun incidents that occur in the specific country where the data was collected - middle-aged, predominantly male, coming from different races, with mixed motives in both public and private locations.

Cluster 1: Young Adults

- Size: 33,301 Incidents (33.0%)
- Mean Age: 24 Years
- **Demographic Profile:** This cluster exhibits large deviations from PBDD, hinting at disproportionate impacts. Firstly, Black representation nearly doubles from 23.1% (PBDD) to 44.1%, while White representation drops from 65.7% (PBDD) to 38%. Hispanic representation also peaks across all clusters, and also increased from 9.0% (PBDD) to 15.1%. Secondly, there is also an increase in the proportion of male to 88.1% (+2.4% from PBDD). Lastly, there is generally an education disadvantage, with more individuals with Less than High School (+12.7% from PBDD), and much lesser Bachelors attainment 3.6% (-9.2% from PBDD). It is worth noting that these young adults may not have been old enough to pursue higher education.
- **Incident Characteristics:** There is a reversal in the type of incidents, with Homicide (61.2%) being dominant, Suicide (35.6%) and Accidental (2.2%, highest across all clusters). Furthermore, Street incidents nearly doubled to 20.7% from 11.1% (PBDD).
- **Interpretation:** Cluster 1 captures racialized urban violence affecting young Black and Hispanic males with limited education attainment. This could reflect systemic inequalities such as failing education systems, young firearm access in socially disorganized communities, gang conflicts or community violence. Thus demographic

deviations reflect the structural gun violence that could be due to failure of government bodies to provide opportunity and a safe environment for these young adults.

Cluster 2: Mature Adults

- Size: 27,429 Incidents (27.2%)
- Mean Age: 57 Years
- **Demographic Profile:** This cluster shows moderate-to-strong deviations, particularly in the racial composition. White representation reaches 86.0% (+20.3% from PBDD), while Black and Hispanic representation drops to 8.1% (-15.0% from PBDD) and 4.2% (-4.8% from PBDD). Female representation peaks highest across all clusters at 17.2%. Educational attainment is generally high, with only 11.8% achieving Less than High School (-9.9% from PBDD).
- **Incident Characteristics:** Strong suicide tendencies, increasing from 62.7% (PBDD) to 81.5%, while homicide incidents drops to 16.3% (-18.6% from PBDD). There is a shift from gun incidents happening at home, which increased to 67.7% (+7.7% from PBDD), and street incidents dropping to 5.0% (-6.1% from PBDD).
- **Interpretation:** Cluster 2 captures adults navigating late-career and early retirement transitions, where they may face retrenchment or start to develop health issues, leading to financial anxieties. The home-centered suicide pattern could suggest that individuals may have planned these acts rather than impulsive violence.

Cluster 3: Elderly

- Size: 14,671 Incidents (14.6%)
- Mean Age: 78 Years
- **Demographic Profile:** This cluster exhibits the largest deviation compared to the PBDD, representing the most distinctive and homogenous group. Race is skewed towards White (93.5%) followed by small groups of other races such as Black and Hispanic (3.3% and 2.4% respectively). Male representation also peaks across all clusters at 89.7%. Education attainment is generally high, with 18.4% less than high school, and highest Bachelor's attainment across all clusters at 20.5% (+7.7% from PBDD). However education does not seem to protect against suicide gun incidents.
- **Incident Characteristics:** There is a high concentration of suicide incidents of 92.1% (+29.4% from PBDD), with much lesser homicide of 6.3% (-9.1% from PBDD) and accidental incidents (1.2%). Location pattern shows majority of the incidents happening at home (80%, +20% from PBDD). Police involvement is at its lowest across all clusters (0.2%, -1.2% from PBDD), which could be due to the nature of suicide gun incidents, which may not require police involvement.
- **Interpretation:** Cluster 3 captures elderly individuals, predominantly White male that could be experiencing the ailments of old age such as terminal illness, chronic pain or cognitive decline, leading to loss of independence. These reasons could result in the elderly choosing to end their life.

5.2.4 Feature Reduction Experiment

The initial K-Prototypes clustering makes use of 35 features (1 numerical and 34 one-hot encoded categorical). While these features captures all the available information, the results of the cluster profile and exploratory data analysis revealed that the following features showed minimal variation across clusters, potentially contributing noise instead of providing meaningful distinctions:

1. **Police Involvement:** Ranged from 98.1% to 99.8% “No” across clusters, with PBDD of 98.6%.
2. **Day of Week:** Showed uniform distribution of about 14.3% per day across all clusters, identical to PBDD.
3. **Is Weekend:** ranged from 71.0% to 71.3% weekday across clusters, compared to 71% PBDD.

To mitigate the issue of Curse of Dimensionality of having too many features, we streamlined the feature set by removing the abovementioned features, and repeated the K-Prototypes clustering.

Comparison: Full features vs Reduced features

Metric	K = 4 Full	K = 4 Reduced	Winner	Interpretation
Features	37	28	-	-
Cost	7426900	6158071	K = 4 Reduced	Lower total dissimilarity
Davies-Bouldin Index	3.6642	3.0411	K = 4 Reduced	Lower indicates better cluster separation
Calinski-Harabasz Index	10834.01	15646.56	K = 4 Reduced	Higher indicates better-defined clusters
Silhouette Score	0.0446	0.0700	K = 4 Reduced	Measures cluster cohesion and separation

Overall, the reduced model performed slightly better in all metrics compared to the full model, which should be expected. While the sizes of each cluster in the reduced model changed slightly, there are minimal changes in the characteristics of each cluster, which further

supports the importance of reduced feature set in providing discriminatory power between the clusters.

5.2.5 Conclusion

While the K-Prototypes clustering algorithm found four distinct age-stratified demographic segments within the guns incident dataset, the evaluation metrics obtained were still considerably poor. For example, a silhouette score of 1 represents perfect clusters, but a score of 0.0446 was obtained for $K = 4$ clusters.

We suggest several future improvements to the model to obtain better clusters:

1. Plotting the evaluation metrics (Davies-Bouldin Index, Calinski-Harabasz Index and Silhouette Score) against the number of clusters (K) to investigate the interactions between them to obtain better-defined segments.
2. Enriching the dataset with additional contextual features such as geographical coordinates (longitude and latitude) to expand onto spatial clustering patterns or socioeconomic indicators for victims/neighbourhood of gun incidents. These additional dimensions could reveal other important factors that may influence gun violence patterns.
3. Further reducing the number of features by employing feature importance as in Section 4.2.7, by removing features of low variation or minimal contribution to the cluster differences, hence reducing the noise that these features contribute, resulting in more distinct clusters.

5.3 Discussion

Both Hierarchical Clustering and K-Prototypes analyses provide complementary perspectives on gun violence, revealing patterns at different levels of granularity. Rather than contradicting each other, it provides a nested hierarchical structure within gun violence incidents. The Hierarchical Clustering ($K=2$) provides a macro division based on intent (homicide-dominant vs suicide-dominant incidents), while the K-Prototypes Clustering ($K=4$) subdivides each of these macro clusters by age, followed by exploring the different demographics and life-stage contexts within each subgroups.

Together, the clustering algorithms create clusters that enable interventions that are appropriately tailored to both the nature of gun incidents and the specific demographic experiencing it.

6. Conclusions

This project investigated gun violence through 3 core research questions:

1. How demographic, situational and contextual factors shape both the reasons for gun incidents and the likelihood of police involvement;
2. Whether underlying behavioural or demographic patterns can be uncovered through clustering; and
3. Whether machine learning models can reliably identify the key risk factors differentiating major categories of gun incidents.

Across supervised learning, unsupervised clustering and the respective metrics evaluation, our analyses reveal convergent insights that speak directly to these questions.

First, in addressing how demographic and situational factors influence incident types and police involvement, our classification models (see 4.1 and 4.2) showed that contextual and location-based variables, such as incidents occurring in the streets, at unspecified locations, or demographic variables such as age and sex played far weaker roles. These findings indicate that gun violence outcomes and police mobilization are shaped less by who is involved and more by where and under what circumstances incidents occur, directly answering our first research question. Error analysis further revealed systemic underprediction in unspecified or incomplete locations (see 4.1), suggesting that limitations in data quality can generate biases in model performance.

Second, our unsupervised clustering analyses provided insights into whether distinct behavioural or demographic groupings exist within the data. Both Hierarchical Clustering (see 5.1) and K-Prototypes (see 5.2) uncovered a clear structural divide, between a suicide/home cluster and a homicide/street cluster, and distinct age-stratified segmentations, revealing coherent situational and demographic patterns. These clusters validate the presence of meaningful behavioural distinctions and show that the patterns identified through clustering align closely with those seen in the classification outcomes. This directly addresses our second key question. However, cluster quality scores remained low, indicating that while broad structural patterns exist, more nuanced segmentation would require richer contextual information such as geospatial or socioeconomic features.

Third, in evaluating whether machine learning models can reliably identify risk factors across suicide, homicide, accidental and undetermined incidents, our classification work revealed a central challenge: extreme class imbalance (see 4.2 and 4.3). While models performed strongly on the dominant categories, they struggled to learn rare outcomes due to their minimal representation. This resulted in high accuracy and micro-averaged scores but poor macro-averaged performance, showing that the models learnt the majority patterns well but could not reliably generalise to underrepresented categories. Thus, the answer to the third research question is qualified: machine learning models can identify key risk factors for well-represented categories, but they cannot be expected to learn reliable signals for rare

forms of gun violence without more balanced data. Finally, our project also highlights that performance metrics can be misleading in heavily imbalanced datasets and highlights a broader theme across our analyses: the core limitations modelling complex, multifaceted social issues like gun violence are statistical, not algorithmic.

Overall, our findings reveal that patterns in gun violence are structural and contextual in nature. The strongest predictive signals relate to the circumstances of the incident rather than demographic traits, and meaningful behavioural clusters do exist but are shaped by context more than by individual characteristics. At the same time, severe data imbalance fundamentally restricts the ability of machine learning models to learn rare incident types, underscoring the need for more complete and representative datasets. These insights collectively highlight that any predictive system built from such data should be used as a decision-support tool, integrated alongside professional expertise, contextual assessment and improved data collection.

7. Appendix A: Statement of Independent Work

1A. Declaration of Original Work. By entering our Student ID below, we certify that we completed our assignment independently of all others (except where sanctioned during in-class sessions), obeying the class policy outlined in the introductory lecture. In particular, we are allowed to discuss the problems and solutions in this assignment, but have waited at least 30 minutes by doing other activities unrelated to class before attempting to complete or modify our answers as per the class policy.

Signed,
Beckham Toh Yuexi (A0265133L)
Iffah Ilyana Binte Hatib (A0258974H)
Ong Si Ying Rean (A0257730A)
Joshua Han Rui Chou (A0272122U)
Tee Peng Chun (A0271711M)

1B. Exception to the Class Policy. We did not follow the CS3244 Class Policy in doing this assignment. This text explains why and how we believe we should be assessed for this assignment given the circumstances explained.

2. References. We give credit where credit is due. We acknowledge that we used the following websites or contacts to complete this assignment:

Carnot. (2024, July 2). Understanding Dimensionality Reduction: PCA vs t-SNE vs UMAP vs Fit-SNE vs LargeVis vs Laplacian Eigenmaps. *Medium*.
<https://carnotresearch.medium.com/understanding-dimensionality-reduction-pca-vs-t-sne-vs-umap-vs-fit-sne-vs-largevis-vs-laplacian-13d0be9ef7f4>

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8. Appendix B: Figures

Figure 1: Age Distribution

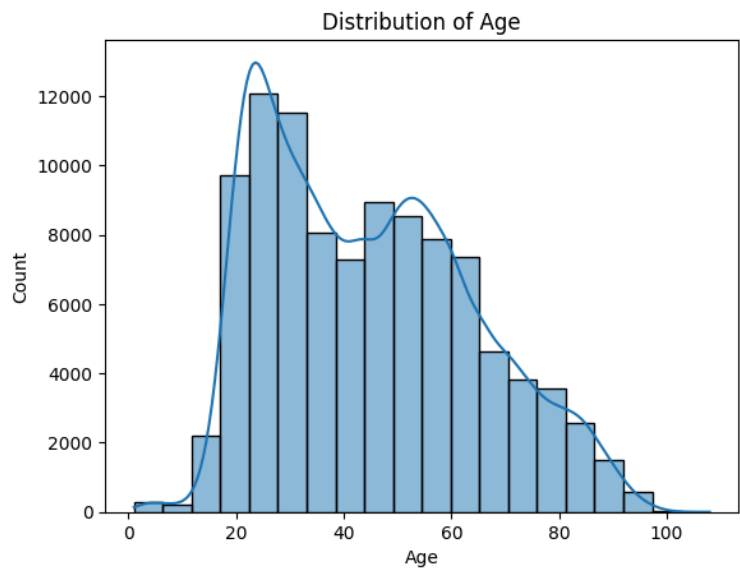


Figure 2: Sex Distribution

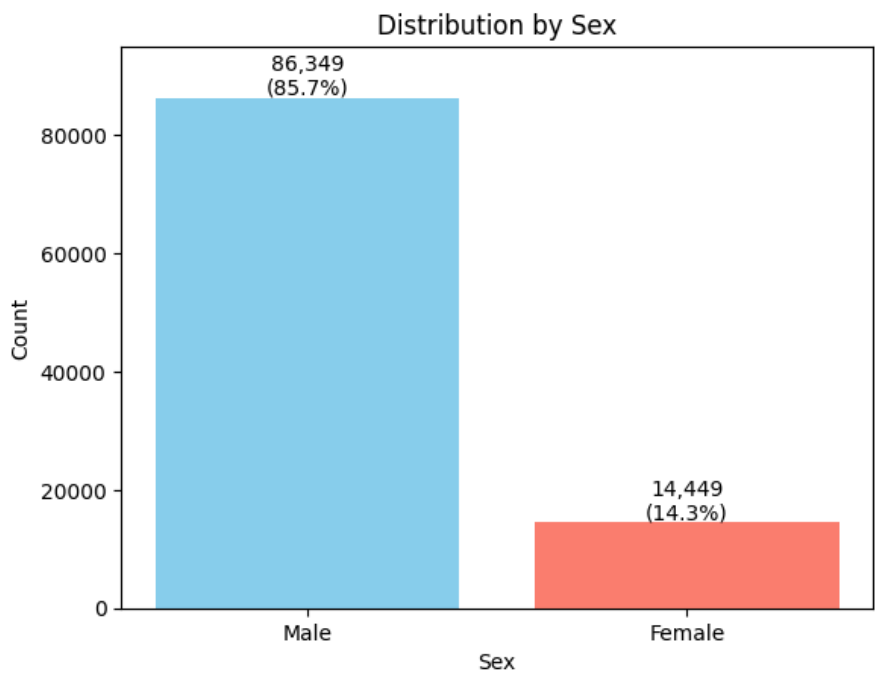


Figure 3: Education Level Distribution

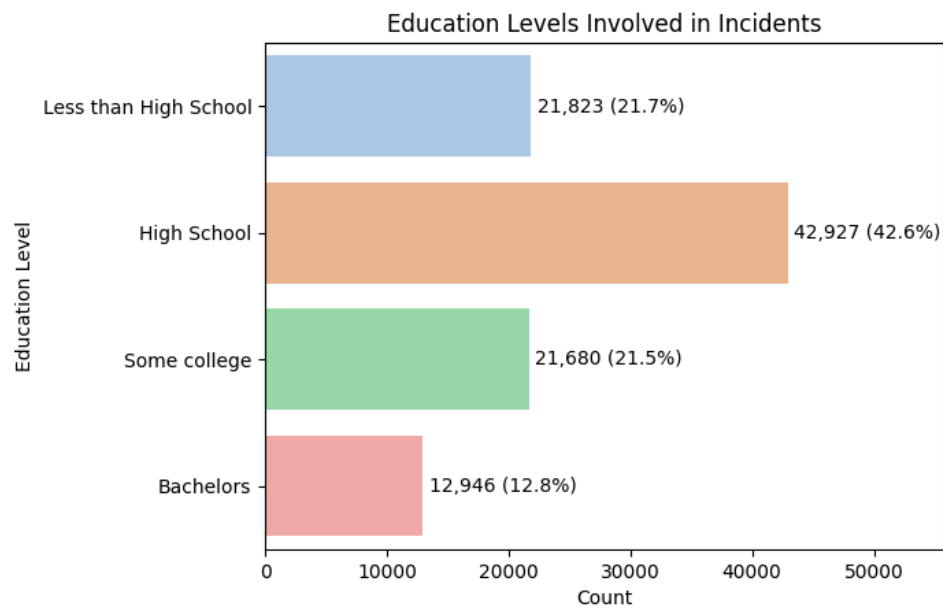


Figure 4: Race Distribution

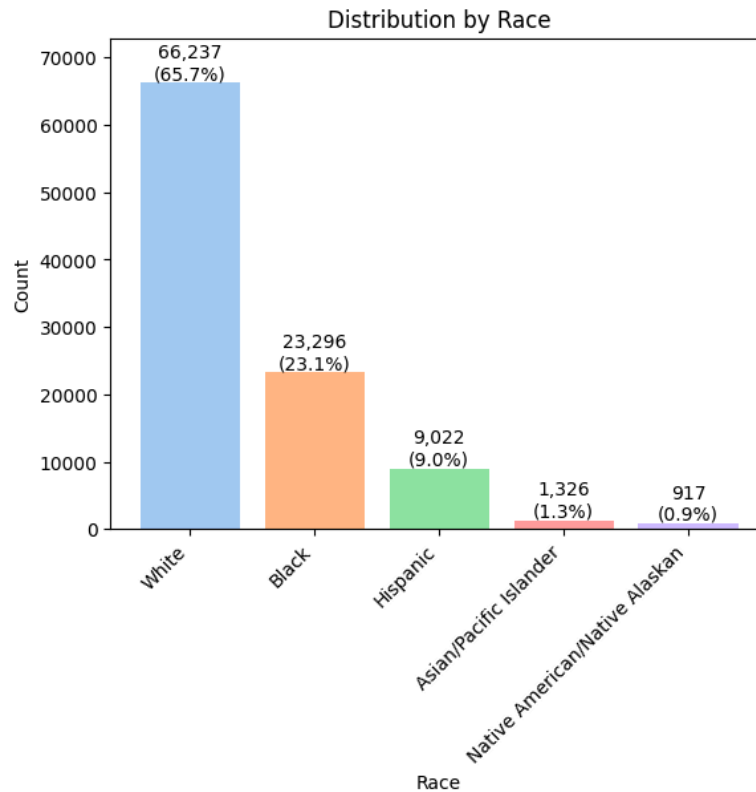


Figure 5: Place of Incident

Place of incident	Count	Percentage (%)
Home	60486	60.0
Other specified	13751	13.6
Street	11151	11.1
Other unspecified	8867	8.8
Trade/service area	3439	3.4
Unspecified	1384	1.4
School/instiution	671	0.7
Farm	470	0.5
Industrial/construction	248	0.2
Residential institution	203	0.2
Sports	128	0.1

Figure 6: Reason for Incident

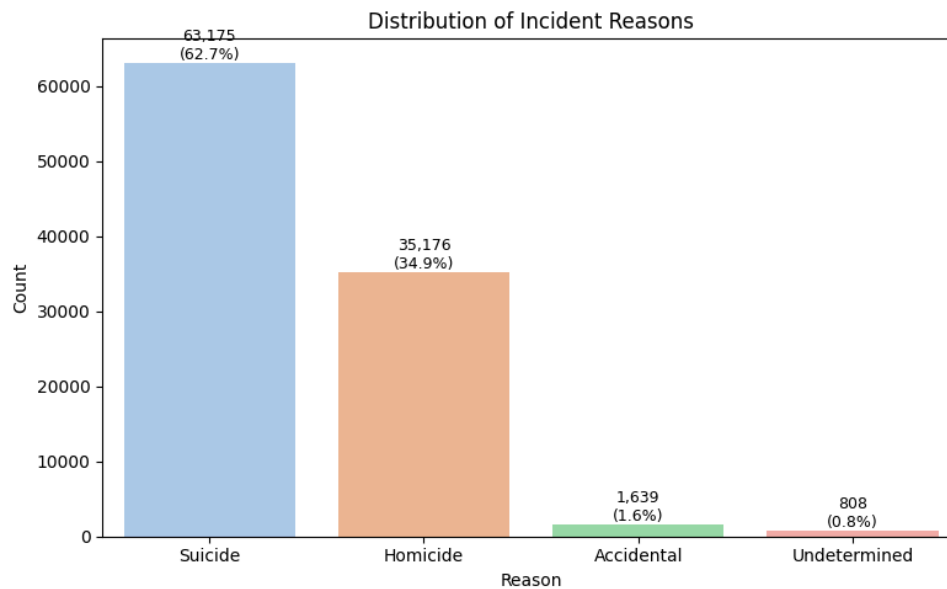


Figure 7: Boxplot on Age distribution by Reason

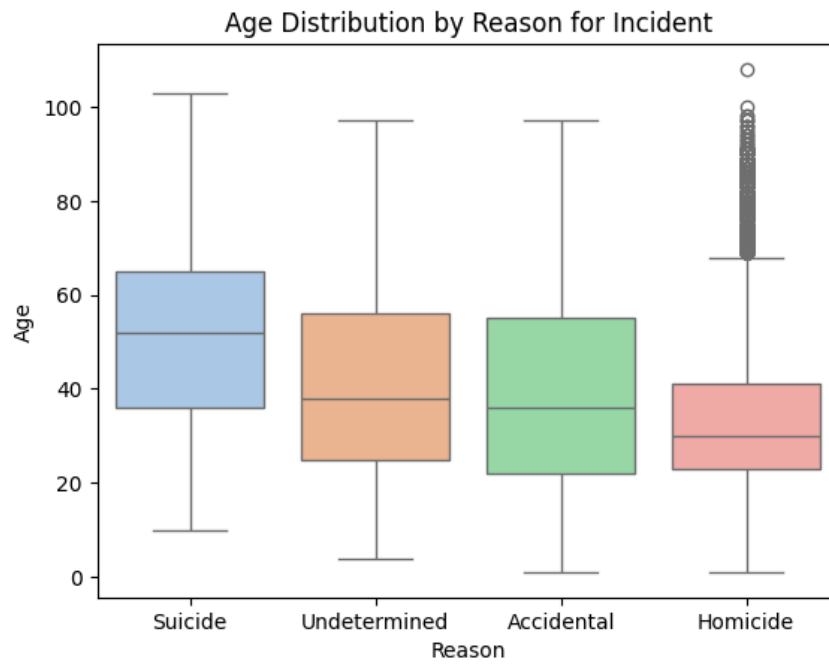


Figure 8: Density plot on Age distribution by Reason

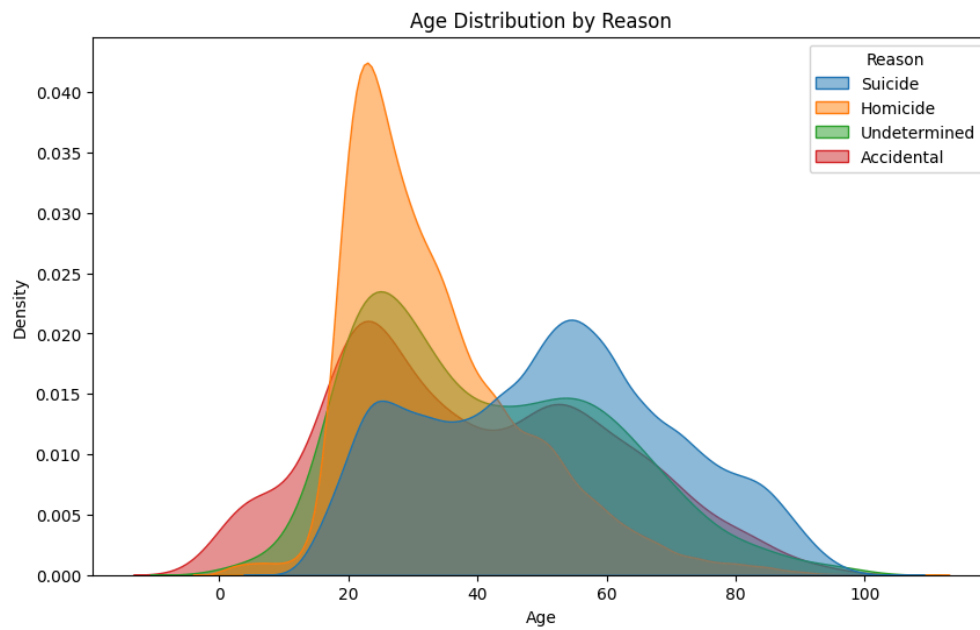


Figure 9: Incident Reason by Sex

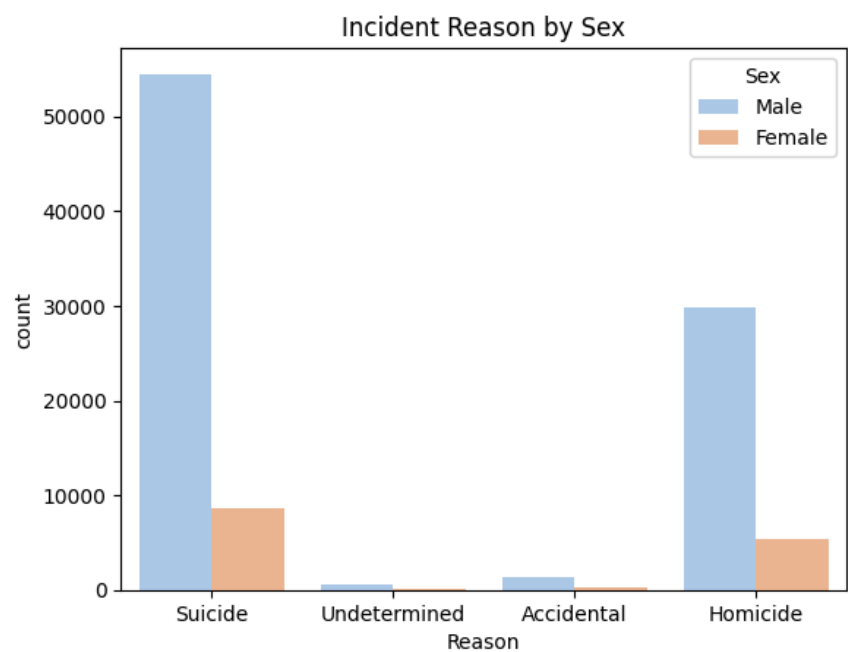


Figure 10: Reason of Incident by Race

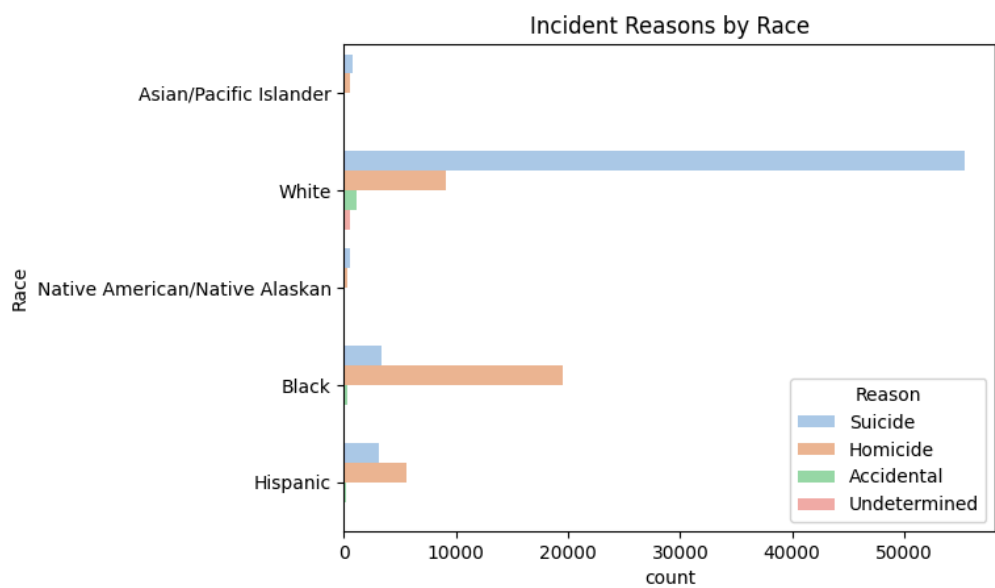


Figure 11: Education Levels by Reason of Incident

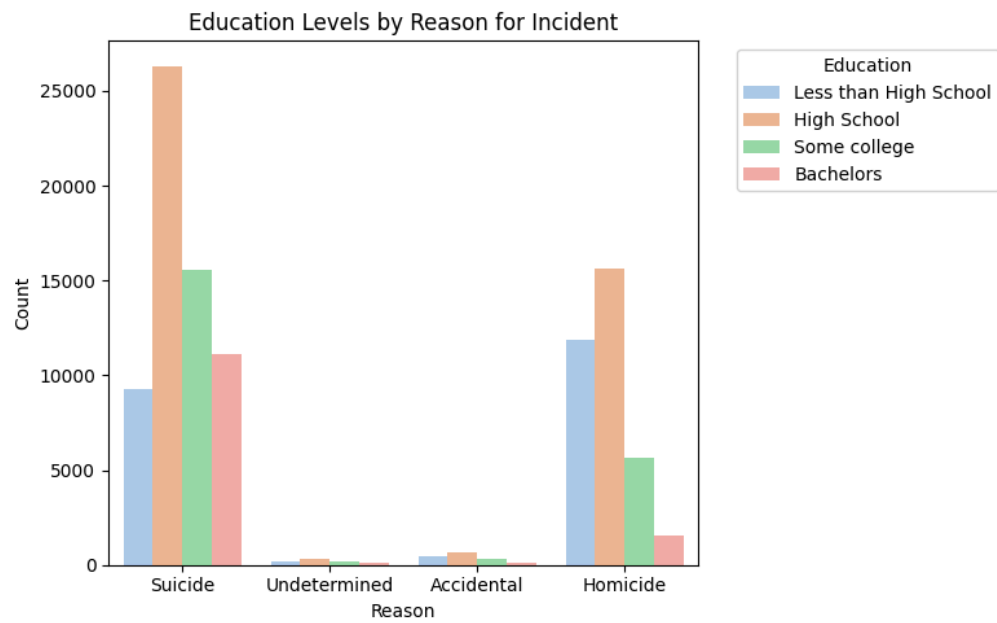


Figure 12: Police Involvement

Police involvement	Count	Percentage (%)
0	99396	98.6
1	1402	1.4

Figure 13: Spread of Incidents across time with 30-day average

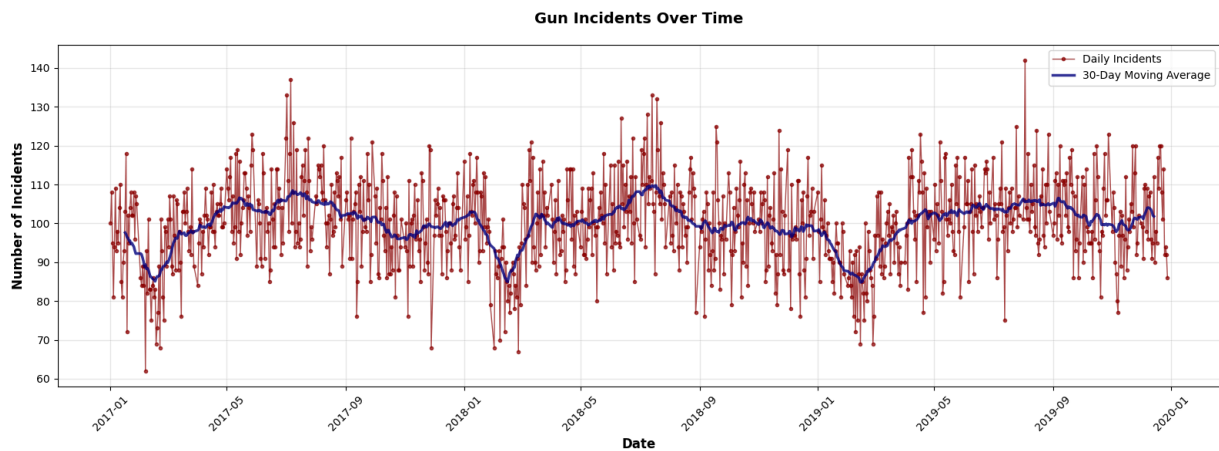


Figure 14: Spread of Incidents across time by months

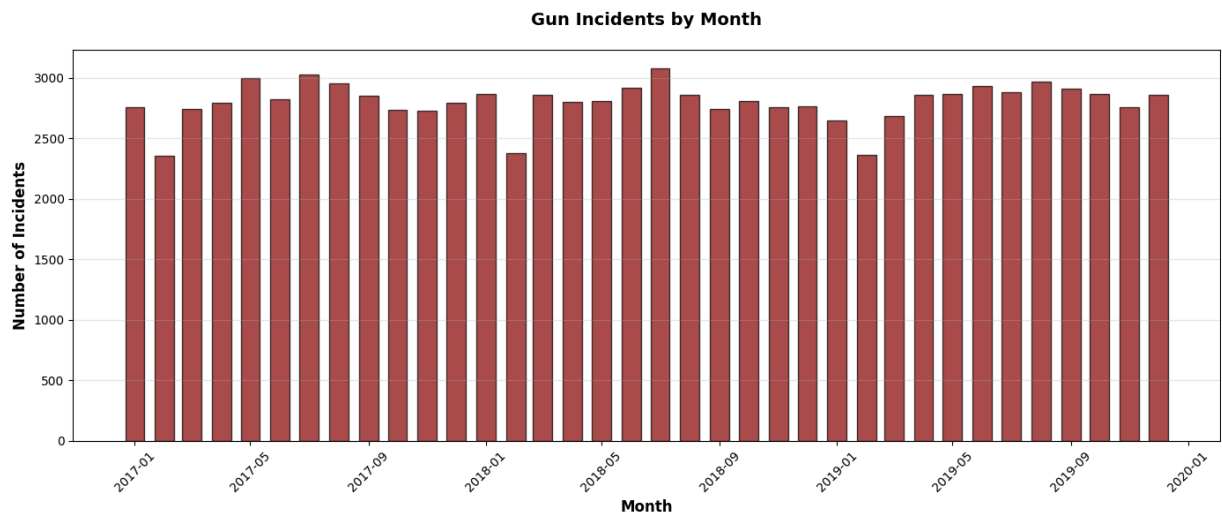


Figure 15: Spread of Incidents across days of the week

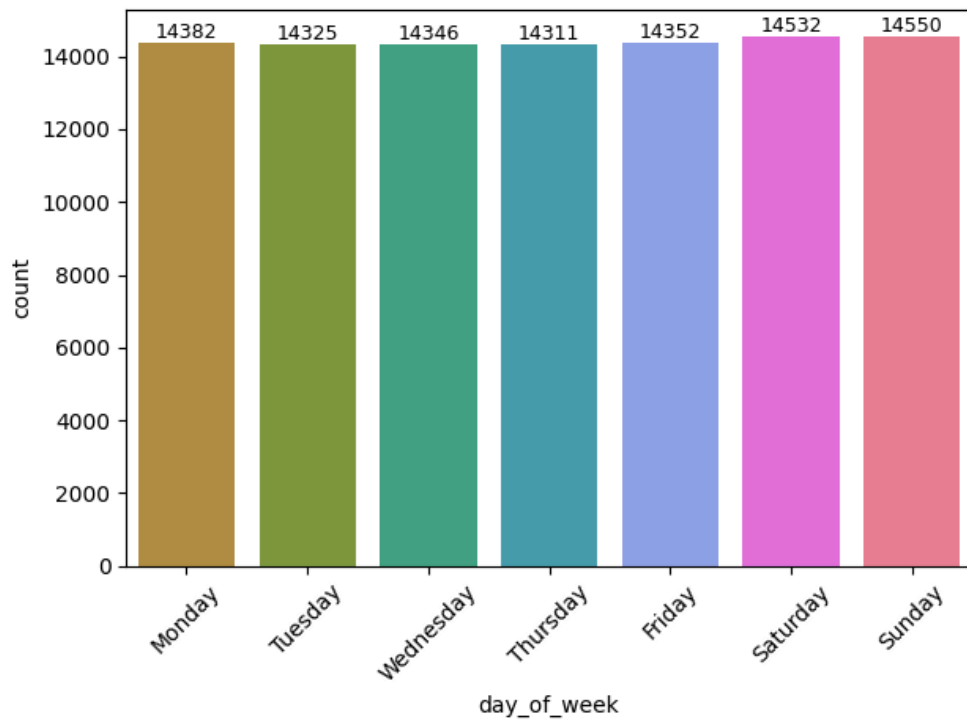


Figure 16: Police Involvement: Police Involvement by Incident

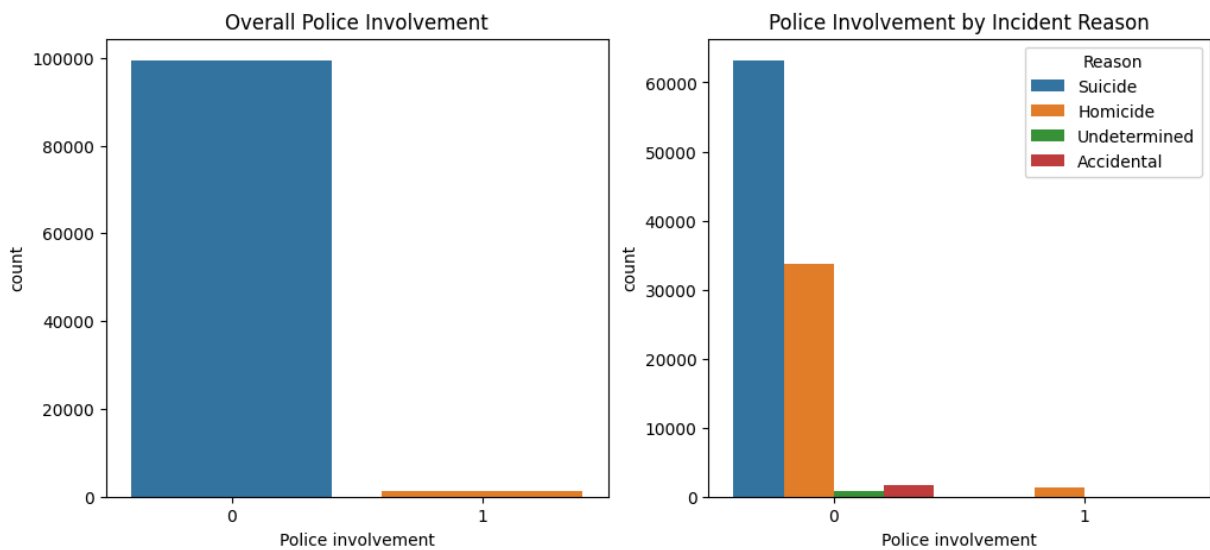


Figure 17: Police Involvement by Sex

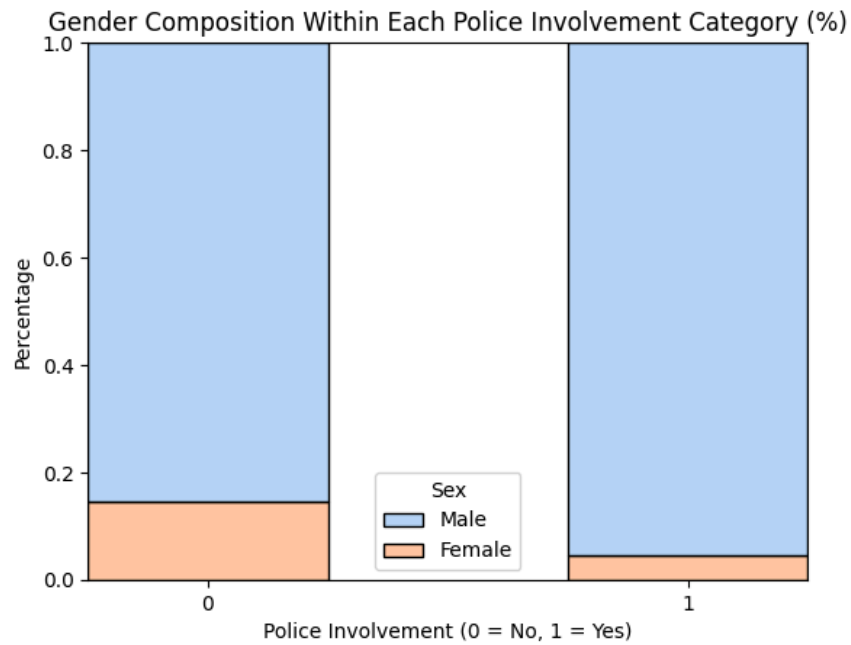


Figure 18: Police Involvement against Place of Incident

Police involvement	0	1	Total	% Police Involved	% No Police
Place of incident					
Home	60483	3	60486	0.0	100.0
Other specified	13749	2	13751	0.0	100.0
Street	11146	5	11151	0.0	100.0
Other unspecified	8859	8	8867	0.1	99.9
Trade/service area	3438	1	3439	0.0	100.0
Unspecified	1	1383	1384	99.9	0.1
School/institution	671	0	671	0.0	100.0
Farm	470	0	470	0.0	100.0
Industrial/construction	248	0	248	0.0	100.0
Residential institution	203	0	203	0.0	100.0
Sports	128	0	128	0.0	100.0